

A Parameterized Training Corpus for the Dirichlet–Neumann Operator

Design, implementation checks, and direct comparison with `combined_dataset_v9`

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Abstract

The historical training archive for the periodic Dirichlet–Neumann operator combines generators with different sampling, filtering, target, and snapshot rules. It is therefore difficult to separate the effects of physical coverage, numerical labels, and source-dependent row counts.

This document specifies a prospective replacement corpus with four physical families: fifth-order Stokes states, periodic Tanaka profiles, Benjamin–Feir sideband states, and resolved-band JONSWAP/TMA random seas. All four use one discrete order-six DNO target, case-level splitting, equal case weight, and reproducible case records. The three trajectory families use one residual-controlled two-stage, fourth-order Gauss–Legendre calculation with step 0.01; a case is retained only when that calculation returns a complete admissible trajectory. Static Stokes cases instead require a finite state in the declared Stokes support. Airy, manufactured-DNO, and sparse-Fourier cases are reserved for validation.

The four samplers, constructors, common target, one-arm trajectory executor, bounded accepted-case scheduler, restart records, combined dataset view, and case-balanced loader are implemented and pass focused CPU tests. The corpus itself has not been generated. Its remaining numerical gate is an exact-contract GPU calculation that obtains one accepted case from each trajectory parameter category.

Contents

1	Dataset organization	5
2	Learning problem and discrete target	5
3	Why regenerate the corpus	7
4	Parameterized initial conditions	7
4.1	Fifth-order Stokes states	8
4.2	Periodic Tanaka-profile sums	10
4.3	Benjamin–Feir carrier and sidebands	12
4.4	JONSWAP/TMA random-phase states	13
4.5	Production parameter sampling	17
4.6	Durable generation and dataset assembly	18
4.7	Training corpus at a glance	18
5	Attempted and accepted counts	21

6	Reference evolution and whole-case acceptance	21
6.1	Discrete Zakharov flow	21
6.2	Residual-controlled Gauss–Legendre stages	22
6.3	Method-level fixed-band resolution validation	23
6.4	Method-level selection of the production time step	24
6.5	Whole-case production acceptance	25
6.6	Diagnostics not used for acceptance	27
6.7	Storage after acceptance	27
7	Reference-method and surrogate validation	27
8	Temporal sampling, splitting, and normalization	28
9	Execution plan	32
10	Claims this plan does and does not support	33
A	Audit of the current v9 archive	34
B	Diagnostic specifications and separation from training	35
B.1	Airy linear-limit panel	35
B.2	Manufactured exact-DNO panel	37
B.3	Sparse-Fourier shallow stress suite	37
C	Rescaling to the 2π reference torus	38

Reading guide. The document proceeds from the DNO target to the four initial-condition formulas, reference rollouts, whole-case acceptance, and balanced training batches. Sections [1](#), [2](#), [4](#), [6](#), and [8](#) define that chain. The difference from the historical archive and the work remaining before generation are stated separately. The detailed **v9** audit is in Appendix [A](#).

Implementation status

The replacement corpus itself has not yet been generated. The following table distinguishes implemented definitions and tests from the one remaining pre-generation calculation:

Item	Status	Evidence
Discrete target	Frozen	$L = 2\pi$, $g = 1$, order six, $N = 1024$, pad factor eight, and delivered band $ k \leq 128$; fixed-band, mean, translation, and dtype tests pass.
Finite-depth Stokes	Exact bounded path passed	The four-category sampler enforces the conservative condition $Ur_+ \leq 26$. At $(N, M, p, K) = (1024, 6, 8, 128)$, one revision-2 case per category passed the complete sampled-specification, construction, target, quality, archive, and dataset-view transaction through the bounded quota executor. The loader returns four finite one-row cases.
Tanaka	Sampler, constructor, and quota path implemented	The eleven-category direction-aware periodic-gap sampler and tangent-Hermite constructor are connected to the common transaction. Only a finite negative potential radicand is outside the declared construction; nonfinite algebra or an invalid wave speed stops generation. A reduced revision-1 wiring smoke reached the dataset loader.
Benjamin–Feir	Sampler and quota path implemented	The 66 admissible integer pairs are parameter categories. Conditional steepness, amplitude ratio, and phase sampling are connected through a durable attempt record to the equation-(33) constructor and common trajectory transaction. A reduced revision-1 wiring smoke reached the dataset loader.
Random sea	Sampler and quota path implemented	The 27 declared categories have continuous parameter laws and stored phase arrays. A reduced revision-1 wiring smoke passed through the durable transaction and dataset view.
Common trajectory transaction	Implemented	One float64 rollout with GL2 step 0.01, saved spacing 0.08, stage residual telemetry, post-acceptance time selection, a complete-case writer, and a split-aware loader are executable and tested. The 0.01/0.005 pair remains a validation calculation and is not run for each production case.
Accepted-quota recovery	Bounded run specification implemented	The driver reconstructs checked accepted counts, replays proposal-only and proposal-plus-shard transaction states, and admits one writer. A category requesting Q_i accepted cases has at most $4Q_i$ durable attempts. Pending work at the ceiling is replayed; a resolved deficient category at the ceiling terminates the family run instead of silently changing its population.
Combined view and loader	Implemented	The view verifies transaction hashes without copying field arrays. In each training epoch the loader selects one stored time from every accepted case; validation uses one fixed time per case.
Time-step validation	Fixed diagnostic panel complete	Ten complete cases agree between GL2 steps 0.01 and 0.005 to at most 1.126337×10^{-5} . The steep Tanaka and equation-(33) Benjamin–Feir calculations become incomplete at nearly the same physical times under both steps. The comparison supports the production step 0.01; it is not a production rejection filter.
Generation gate	GPU calculation running	A prior generator-revision-1 paired-step four-family smoke accepted one case per family and assembled 417 finite rows. The revision-2 104-category gate is running. Bulk generation still waits for it to obtain a single-arm production case from each of the 11 Tanaka, 66 Benjamin–Feir, and 27 random-sea parameter categories under the final bounded scheduler.

1 Dataset organization

Let $L > 0$ be the horizontal period and $\mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$ the periodic interval, with horizontal coordinate $x \in \mathbb{T}_L$. Each training case stores a real surface elevation $\eta_0(x)$, a real surface-potential trace $\xi_0(x)$, and a constant mean depth $h > 0$. It also stores whether it was generated as a Stokes, Tanaka, Benjamin–Feir, or random-sea case and every numerical parameter required to reconstruct that initial state. The four constructions are defined explicitly in Section 4. Every accepted initial state satisfies

$$\int_{\mathbb{T}_L} \xi_0(x) \, dx = 0, \quad \min_{x \in \mathbb{T}_L} \{h + \eta_0(x)\} > 0. \quad (1)$$

The first condition fixes the irrelevant additive constant in ξ_0 ; the second excludes an initial dry point. Depth is stored with the case and held fixed during a rollout.

The dataset rules are:

1. Generate the same accepted case count from each family. Balance the coarse parameter ranges listed in Section 4.
2. First require the sampled specification to lie in its declared family support and its declared constructor to return a well-defined periodic initial state. For Tanaka cases specifically, a finite negative radicand is outside the declared construction; a nonfinite radicand or an invalid wave speed stops generation. The trajectory adapters also require the assembled initial arrays to be finite, the depth to be positive, and $h + \eta_0 > 0$ on the grid. Failure of these constructor invariants stops generation rather than becoming an ordinary zero-row draw. For a trajectory family, retain the complete case only if the single fixed-step calculation produces a complete admissible numerical trajectory as defined in Section 6; static Stokes cases retain one well-defined state in their declared support. A failed trajectory contributes zero frames.
3. At each training epoch, choose one stored time uniformly from every accepted training case and then batch the selected rows. Because the accepted case count is the same in all four families, every family and every case has equal epoch weight. Validation uses one fixed, deterministic stored time from every validation case.
4. Keep Airy, manufactured-DNO, and sparse-Fourier cases out of training. They are diagnostic tests only.

There is no production rejection rule based on sign changes, slope, extrema, Hamiltonian drift, absolute target size, or visual appearance. The replacement corpus uses low-steepness Stokes states instead of a separate single-mode family, and a standard finite-depth random-sea spectrum instead of the old Gaussian sea. Its shallow mixed-mode coverage is tested as specified in Section 9. The existing **v9** archive remains the experimental baseline.

2 Learning problem and discrete target

Let $g > 0$ denote gravitational acceleration; the generated corpus uses the nondimensional value $g = 1$. For each integer Fourier index n , define the wavenumber $k_n = 2\pi n/L$. For any integrable L -periodic function u , use the Fourier convention

$$\hat{u}_n = \frac{1}{L} \int_{\mathbb{T}_L} u(x) e^{-ik_n x} \, dx, \quad u(x) = \sum_{n \in \mathbb{Z}} \hat{u}_n e^{ik_n x}. \quad (2)$$

For an N -point grid, define the spacing $\Delta x = L/N$ and nodes $x_j = j\Delta x$ for $j = 0, \dots, N-1$. The normalized continuous and grid norms are

$$\|u\|_{2,L}^2 = \frac{1}{L} \int_{\mathbb{T}_L} |u|^2 dx, \quad \|u\|_{2,N}^2 = \frac{1}{N} \sum_{j=0}^{N-1} |u(x_j)|^2. \quad (3)$$

Unless a subscript is shown, $\|\cdot\|_2$ means the corresponding normalized norm on the representation being discussed. Fix a maximum retained wavenumber $K > 0$. Let P_K be the fixed-band projection, let $\mathcal{J}_K$ be the set of its retained positive Fourier indices, and let the indicator $\mathbf{1}_{\{\cdot\}}$ equal one when the condition in braces holds and zero otherwise:

$$\widehat{P_K u_n} = \mathbf{1}_{\{|k_n| \leq K\}} \widehat{u_n}, \quad \mathcal{J}_K = \{n \in \mathbb{Z} : n \geq 1, k_n \leq K\}. \quad (4)$$

The projection acts componentwise when its argument is an ordered pair of fields. Define the mean-removal operator $P_0^\perp$ and the mean-zero band projection P_K° by

$$P_0^\perp u = u - \widehat{u}_0, \quad P_K^\circ = P_0^\perp P_K. \quad (5)$$

Thus P_K° both restricts the band and fixes the zero-mean gauge. It is used for DNO outputs and for the ξ component of the vector field; P_K alone is used for η , whose mean is the conserved mass. The frozen target cutoff is $K = 128$, and the random-sea transition begins at $K_0 = 96$. For a nonnegative wavenumber k , the random-sea formula uses the separate smooth window

$$W_{K,K_0}(k) = \begin{cases} 1, & 0 \leq k \leq K_0, \\ \cos^2 \left[\frac{\pi}{2} \frac{k - K_0}{K - K_0} \right], & K_0 < k < K, \\ 0, & k \geq K. \end{cases} \quad (6)$$

Unlike P_K , this window changes the constructed initial condition; the two operations must not be conflated. We write $W_K := W_{K,K_0}$ for the frozen choice $(K_0, K) = (96, 128)$.

For a surface elevation $\eta(x)$ and depth $h > 0$, let y denote the vertical coordinate and define the fluid domain

$$\Omega_{\eta,h} = \{(x, y) : x \in \mathbb{T}_L, -h < y < \eta(x)\}. \quad (7)$$

For surface Dirichlet data ξ , let φ solve

$$\Delta \varphi = 0 \quad \text{in } \Omega_{\eta,h}, \quad \varphi_y(x, -h) = 0, \quad \varphi(x, \eta(x)) = \xi(x). \quad (8)$$

The physical Dirichlet–Neumann operator is

$$G(\eta; h)\xi = (\varphi_y - \eta_x \varphi_x)_{y=\eta(x)}. \quad (9)$$

Let $D = -i\partial_x$, so $|D|$ is the Fourier multiplier with symbol $|k_n|$. For each integer $m \geq 0$, let G_m be the Craig–Sulem term homogeneous of degree m in η , and let M be the largest retained order. The current solver uses

$$G_M(\eta; h)\xi = \sum_{m=0}^M G_m(\eta; h)\xi, \quad G_0(h) = |D| \tanh(h|D|). \quad (10)$$

In particular,

$$G_1(\eta; h)\xi = -G_0(h)(\eta G_0(h)\xi) - \partial_x(\eta \partial_x \xi). \quad (11)$$

This formula fixes the meaning of the analytic $G_0 + G_1$ backbone referenced later in the diagnostic discussion.

We distinguish the physical operator in (9) from its fully discrete evaluation. Let $\mathcal{G}_M^{N,p}$ denote the order- M pseudospectral recursion on N points with padding factor p . The frozen common learning target is

$$\boxed{G_{\text{ref}}(\eta; h)\xi = P_K^\circ \left[\mathcal{G}_M^{N,p}(P_K\eta; h)(P_K\xi) \right]}. \quad (12)$$

The frozen paper target uses

$$L = 2\pi, \quad g = 1, \quad M = 6, \quad N = 1024, \quad p = 8, \quad K = 128. \quad (13)$$

The evaluation in (12) is performed in float64 and only the final arrays are stored in float32. Statements that a constructed field has zero mean refer to the float64 value before storage; float32 conversion preserves the condition only to rounding error.

Equation (12) says exactly which discrete formula the model is trained to reproduce. A claim that this formula approximates the physical G requires separate convergence evidence in M , N , and p . Fixed- M agreement between several grids does not measure Craig–Sulem truncation error. The implementation also tests that unresolved input modes do not change any delivered field, that $G_{\text{ref}}(\eta; h)\xi$ has zero mean, and that the target commutes with translations on the periodic grid. The manuscript therefore claims approximation of this declared discrete order-six target, not a measured continuum-DNO error.

For an accepted case, let t denote time and write the reference state as $z(t) = (\eta(t), \xi(t), h)$. Each stored training example is

$$(P_K\eta(t), P_K\xi(t), h, G_{\text{ref}}(\eta(t); h)\xi(t)). \quad (14)$$

The complete case, rather than the individual stored example, is the unit used for rejection and train/validation/test splitting.

3 Why regenerate the corpus

The current v9 archive is a useful performance baseline, but its sources do not share one data rule. They contribute different numbers of snapshots per physical case, use different output filtering and rejection rules, and do not always store enough information to reconstruct a rejected attempt. Consequently, a change in performance cannot be attributed cleanly to the initial-condition family alone.

The replacement corpus is defined in terms of complete case records. Every family uses the same target, case-level split, and batch weighting. Tanaka, Benjamin–Feir, and random-sea trajectories use the same reference solver and the same complete-trajectory requirement. Static Stokes cases instead use their declared parameter support, the frozen target, and generator-level spatial validation; they are not assigned a fictitious time-integration gate. Appendix A gives the exact v9 row counts, historical parameter ranges, and formulas needed for comparison; none of those old source names is treated as a new physical family.

4 Parameterized initial conditions

This section gives an explicit initial-state formula and parameter list for each family. The complete parameter list is written before integration, for both accepted and rejected attempts. Numerical

formula versions, L , g , N , spectral restriction, and solver tolerances belong to a common construction configuration rather than being repeated in every case record. Here $\text{Unif}[a, b]$ denotes the uniform distribution on $[a, b]$. A positive variable is log-uniform on $[a, b]$ when its logarithm is uniform on $[\log a, \log b]$. The notation $\text{Dirichlet}(1, \dots, 1)$ denotes the uniform distribution on the simplex $\{(w_1, \dots, w_m) : w_j \geq 0, \sum_j w_j = 1\}$.

4.1 Fifth-order Stokes states

Choose an integer carrier mode $n \geq 1$, amplitude parameter $a > 0$, phase $\phi \in [0, 2\pi)$, depth $h > 0$, and branch $r \in \{\text{finite}, \text{deep}\}$. Set $k = k_n$ and define the steepness $\epsilon = ka$. Let $\mathcal{S}_5$ denote the deterministic fifth-order Stokes constructor used by the project. It evaluates the frozen finite-depth coefficient table when $r = \text{finite}$ and its deep-water limit when $r = \text{deep}$; the fifth-order expansion and its range of application are discussed by Fenton and by Zhao and Liu [2, 1]. Define

$$(\eta_0, \xi_0) = \mathcal{S}_5(h, n, a, \phi, r). \quad (15)$$

The implemented generator contains the full coefficient table and its version identifier. Each Stokes case records (h, n, a, ϕ, r) and that formula version. The mean of ξ_0 is fixed to zero because adding an x -constant does not change the DNO. Finite and deep are parameter strata, not different physical families. The JAX constructor has also been checked against 21 snapshots produced independently by the collaborator’s MATLAB generator: relative discrepancies are at most 1.1×10^{-14} in η , 1.0×10^{-14} in ξ , and 9.0×10^{-13} in the order-six DNO output. Thus the shallow-support restriction below addresses the range of the perturbation formula, not a MATLAB-to-JAX transcription error. The sampling population has four parameter categories: the Cartesian product of the finite and deep branches with the two steepness intervals

$$0.005 \leq ka < 0.03 \quad \text{and} \quad 0.03 \leq ka \leq 0.15, \quad (16)$$

with the common amplitude interval $a \in [0.000766, 0.011494]$. On the finite branch,

$$n \in \{14, \dots, 26\}, \quad 0.02 \leq h \leq 1.5, \quad 0.5 \leq kh \leq 5.$$

On the deep branch,

$$n \in \{1, \dots, 20\}, \quad 4 \leq h \leq 50, \quad kh \geq 5.$$

Within a preassigned branch–steepness category, first list the integer modes for which the depth, amplitude, and steepness intervals have nonempty intersection. Choose uniformly from that finite list. Conditional on the mode, choose h log-uniformly on its allowed depth interval, choose ϕ uniformly on $[0, 2\pi)$, and choose a uniformly on the intersection of the common amplitude interval with the assigned steepness interval. Only feasible modes enter the list, and an attempted case never changes its preassigned branch or steepness category.

This population omits both the small fraction of archived deep-Stokes states with $ka < 0.005$ and the $n = 1$, $4 \leq h < 5$ corner rather than claiming that it preserves the entire historical support. The low- ka category is part of the same Stokes family and formula limit, not a fifth training family. The fifth-order state approaches its Airy limit as $a \rightarrow 0$. Moreover, the paper model evaluates $G_0 + G_1$ analytically and constrains its learned remainder to begin at $O(\eta^2)$; duplicating that limit with hundreds of thousands of standalone linear states is therefore not the preferred use of the label budget. The Airy limit is tested independently in Section B.

The finite-depth coefficient table is a perturbation expansion and is not uniformly accurate over every pair (kh, ka) in the rectangular historical sampling range. Let

$$H = \max_{x \in \mathbb{T}_L} \eta_0(x) - \min_{x \in \mathbb{T}_L} \eta_0(x) \quad (17)$$

be the physical crest-to-trough height of the constructed profile, and let $\lambda = 2\pi/k = L/n$ be its carrier wavelength. Define the Ursell number

$$\text{Ur} = \frac{H\lambda^2}{h^3}. \quad (18)$$

The implemented series is $\eta_0(x) = \sum_{m=1}^5 E_m \cos(m\theta)$, where $\theta = kx + \phi$ and E_m is the coefficient of the m th elevation harmonic. Let H_{4096} be the largest sampled elevation minus the smallest sampled elevation on 4096 equally spaced phases, and define

$$B_2 = \sum_{m=1}^5 m^2 |E_m|, \quad H_+ = H_{4096} + \frac{B_2}{4} \left(\frac{2\pi}{4096} \right)^2. \quad (19)$$

Since $|\text{d}^2\eta_0/\text{d}\theta^2| \leq B_2$, Taylor's theorem in the phase coordinate at the true maximum and minimum gives $H \leq H_+$. Define the conservative value $\text{Ur}_+ = H_+\lambda^2/h^3$. For the finite-depth branch, the sampling domain requires

$$\boxed{\text{Ur}_+ \leq 26}. \quad (20)$$

Fenton identifies $\epsilon/(kh)^3$ as the effective shallow-water expansion parameter for Stokes theory [2]. Zhao, Wang, and Liu use $\text{Ur} = 26$ to delimit the fifth-order Stokes regime from the shallower cnoidal regime [3]. Since the constructed wave height satisfies $H = 2a[1 + O(\epsilon^2)]$,

$$\text{Ur} = 8\pi^2 \frac{ka}{(kh)^3} [1 + O((ka)^2)]. \quad (21)$$

Thus (20) reduces at leading order to $ka \lesssim 0.32929(kh)^3$. The implemented decision uses the upper bound (19), not this leading approximation. It is evaluated from the analytic initial profile before integration and is independent of the sampled phase. The phase ϕ is sampled directly and held fixed; it is not inferred from a time whose nonlinear frequency would change when a changes. The constructor removes the spatial mean of ξ_0 , which fixes the irrelevant constant-potential gauge. A rejected amplitude is resampled within its preassigned category; it is not clipped to the boundary. After the initial amplitude draw, at most 1000 additional amplitude redraws are allowed. Every rejected amplitude and its corresponding Ur_+ value are stored with the accepted case after a successful redraw. If the allowed redraws are exhausted, the ledger instead records the fixed mode, depth, phase, and category, together with every attempted amplitude and Ur_+ value. The executor commits that record as a zero-row out-of-support attempt and schedules a replacement in the same category. The family run stops only if the outer attempt ceiling in (54) is exhausted before the accepted-case quota is filled.

An independently seeded 256-case finite-depth panel contained four nonfinite reference trajectories and three finite trajectories whose maximum recorded relative Hamiltonian drift exceeded 10^{-3} . All seven satisfy $80.50 \leq \text{Ur}_+ \leq 161.82$. The support condition also excludes 18 of the other 249 trajectories. This comparison diagnoses the sampled Stokes support; it does not turn the Ursell bound into a classifier of numerical failure: a general initial state can evolve finitely even when its intended fifth-order traveling-wave interpretation is outside the recommended regime. Historical case 98 has conservative $\text{Ur}_+ = 96.88$ and leading approximation 61.87, so both identify the sharp profile as an unsupported parameter draw. The previously considered ratio of correction-harmonic amplitudes to the fundamental remains a descriptive diagnostic only; no cited Stokes-wave analysis recommends it as a support boundary.

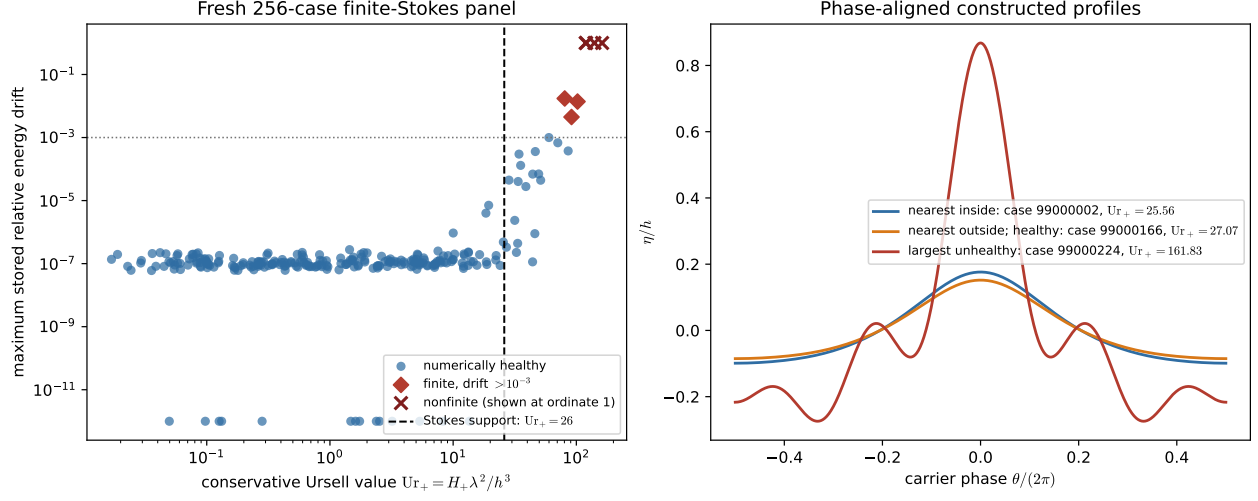


Figure 1: Finite-depth fifth-order Stokes diagnosis. The left panel places independently generated cases against the support boundary $Ur_+ = 26$ and distinguishes nonfinite trajectories and finite trajectories whose recorded relative Hamiltonian drift exceeds 10^{-3} . The right panels show the nearest cases on either side of the boundary and the largest flagged profile. The vertical line defines the adopted family support; it is not fitted as an exact classifier of integration failure.

Any enlargement beyond the current Stokes amplitude and depth support is a separate experiment. In particular, the old manufactured single-mode support ($n = 1, \dots, 20$ and kh as low as 0.02) is much broader than the finite-depth Stokes support and is not silently relabeled as Stokes data.

This is a fifth-order analytic approximation. It must not be described as the numerically exact Fenton Stokes construction used in the long-time tests of Xu and Guyenne [4].

4.2 Periodic Tanaka-profile sums

Let $\alpha = a/h$ be the ratio of crest amplitude a to depth h , and let X denote the dimensionless horizontal coordinate. Following Tanaka's solitary-wave formulation [5], the frozen unit-depth solve at amplitude ratio α returns a dimensionless elevation $\bar{\eta}_\alpha(X)$, a tabulated surface angle $\theta_\alpha(X)$, and a Froude number F_α . Xu and Guyenne likewise use a solitary wave computed by Tanaka's method [4]. The periodic embedding, tangent-Hermite interpolation, multi-profile superposition, and sampling laws below are choices of the present corpus rather than parts of either cited solitary-wave construction. The profile and angle satisfy

$$\frac{d\bar{\eta}_\alpha}{dX}(X) = \tan \theta_\alpha(X). \quad (22)$$

Before interpolation, the constructor checks that the two endpoint elevations and the endpoint and crest tangents differ from zero by at most 10^{-12} , and then sets those values exactly to zero. It forms a cubic Hermite interpolant from the corrected elevations and tangents and extends that interpolant by zero outside its tabulated interval. The value and first derivative therefore join continuously to the zero exterior. For a center x_0 , define its periodic embedding by

$$\eta_{\alpha,h,x_0}(x) = h \sum_{r \in \mathbb{Z}} \bar{\eta}_\alpha \left(\frac{x - x_0 + rL}{h} \right). \quad (23)$$

Because the extended profile has compact support, only finitely many terms are nonzero on one period. The implementation includes every translated copy whose support intersects $[0, L)$. Each copy is evaluated by the cubic Hermite interpolation just defined. The scaling is dimensionally consistent because

$$\partial_x \left[h \bar{\eta}_\alpha \left(\frac{x - x_0 + rL}{h} \right) \right] = \frac{d\bar{\eta}_\alpha}{dX} \left(\frac{x - x_0 + rL}{h} \right). \quad (24)$$

The sum is evaluated on an eight-times finer grid and then restricted to the native Fourier grid.

Set $c_\alpha = F_\alpha \sqrt{gh}$. For direction $s \in \{-1, 1\}$, the traveling-wave formula gives the candidate derivative

$$r_{\alpha, h, x_0, s} = s \left[c_\alpha - \sqrt{(1 + (\partial_x \eta_{\alpha, h, x_0})^2)(c_\alpha^2 - 2g\eta_{\alpha, h, x_0})} \right]. \quad (25)$$

Define its radicand by

$$\mathcal{R}_{\alpha, h, x_0}(x) = (1 + (\partial_x \eta_{\alpha, h, x_0}(x))^2) (c_\alpha^2 - 2g\eta_{\alpha, h, x_0}(x)). \quad (26)$$

The production constructor first requires c_α^2 to be finite and strictly positive and $\mathcal{R}_{\alpha, h, x_0}$ to be finite at every grid point. Failure of either requirement stops generation and is not relabeled as a support rejection. When these quantities are finite, a component with $\min_x \mathcal{R}_{\alpha, h, x_0}(x) < 0$ lies outside the declared Tanaka construction and is recorded as a zero-row case. The radicand is never modified by clipping.

Write $\hat{r}_n$ for the n th Fourier coefficient of $r_{\alpha, h, x_0, s}$. A periodic antiderivative exists only after removing the mean, so the potential is defined by

$$\partial_x \xi_{\alpha, h, x_0, s} = P_0^\perp r_{\alpha, h, x_0, s}, \quad \widehat{\xi_{\alpha, h, x_0, s_n}} = \frac{\hat{r}_n}{ik_n} \quad (n \neq 0), \quad \widehat{\xi_{\alpha, h, x_0, s_0}} = 0. \quad (27)$$

Thus (25) is the unperiodized traveling-wave relation, while (27) is the actual periodic formula. Their difference is the constant $\hat{r}_0$ and is recorded in the case audit.

For an integer component count $m \geq 1$, scaling, periodic embedding, and summation therefore give

$$(\eta_0, \xi_0) = P_K \sum_{j=1}^m (\eta_{\alpha_j, h, x_j}, \xi_{\alpha_j, h, x_j, s_j}). \quad (28)$$

Each Tanaka case records the depth h , component count m , and the amplitude ratio, center, and direction (α_j, x_j, s_j) of every component. The three main structural groups have $m = 1, 2, 3$ respectively and use

$$U_h \sim \text{Unif}[0, 1], \quad h = (0.01)^{1-U_h} (0.30)^{U_h}, \quad A = \sum_{j=1}^m \alpha_j \sim \text{Unif}[0.10, 0.35]. \quad (29)$$

This is geometric interpolation between the endpoint depths. At fixed amplitude ratio, (23) shows that multiplying h by a constant multiplies both the horizontal and vertical scales of a component by that constant. Equal increments of U_h therefore represent equal multiplicative changes of profile scale. This is a numerical coverage rule and is not asserted to describe the frequency of physical water depths. For $m > 1$, draw $(w_1^{(\alpha)}, \dots, w_m^{(\alpha)}) \sim \text{Dirichlet}(1, \dots, 1)$ and set $\alpha_j = Aw_j^{(\alpha)}$; for $m = 1$, set $\alpha_1 = A$. The steep structural group has $m = 1$ and uses

$$U_h \sim \text{Unif}[0, 1], \quad h = (0.20)^{1-U_h} (0.35)^{U_h}, \quad \alpha_1 \sim \text{Unif}[0.25, 0.45]. \quad (30)$$

Let

$$q = \#\{j : s_j = +1\} \quad (31)$$

be the number of right-moving crests. For the main groups, the pairs (m, q) with $m \in \{1, 2, 3\}$ and $q = 0, \dots, m$ define $2 + 3 + 4 = 9$ parameter categories. The steep group has the two categories $q = 0, 1$. Hence the Tanaka family has eleven parameter categories. Conditional on q , the sampler chooses uniformly which q of the m exchangeable crest labels have sign $+1$; all remaining signs are -1 . The accepted-case quota is divided equally over the eleven categories, up to one case when the requested total is not divisible by eleven. Thus left and right directions remain balanced marginally in the ideal equal-category population, but the direction compositions $q = 0, \dots, m$ have equal coverage instead of the binomial frequencies produced by independent fair signs. A finite quotient-remainder assignment may introduce a one-case direction imbalance; with the fixed category order, a quota of 1024 has one extra left-moving one-crest case. This prospective law deliberately gives aggregate weights $2/11$, $3/11$, $4/11$, and $2/11$ to main $m = 1$, main $m = 2$, main $m = 3$, and steep $m = 1$, respectively. It replaces the old four-group independent-sign mixture; the replacement corpus has not yet been generated.

The centers are sampled without privileging the endpoints of a plotting interval. Draw

$$(w_1^{(x)}, \dots, w_m^{(x)}) \sim \text{Dirichlet}(1, \dots, 1), \quad g_j = 3h + (L - 3mh)w_j^{(x)}, \quad (32)$$

use the g_j as the cyclic gaps between consecutive centers, and rotate all centers by one common uniform draw on $[0, L)$. Since $\sum_j g_j = L$, this defines a translation-invariant periodic configuration. The periodic distance $d_L(x_i, x_j) = \min_{r \in \mathbb{Z}} |x_i - x_j + rL|$ therefore satisfies

$$d_L(x_i, x_j) \geq 3h \quad (i \neq j). \quad (33)$$

For each α_j , the numerical profile solve returns one tabulated profile. Its tolerances and coefficient version are fixed for the entire dataset. Periodic continuation is part of the Tanaka construction, not a repair applied after sampling.

A sum of solitary profiles is a constructed initial state, not an exact multi-soliton solution. The eleven categories above belong to the same Tanaka family.

4.3 Benjamin–Feir carrier and sidebands

This family is deep-water only. Choose an integer carrier mode n_c , a positive integer sideband separation $\Delta n < n_c$, a carrier steepness ε_c , a sideband-to-carrier amplitude ratio ρ , and one common sideband phase $\phi \in \mathbb{R}/(2\pi\mathbb{Z})$. The carrier phase is fixed to zero as a translation gauge. Define

$$k_c = \frac{2\pi n_c}{L}, \quad k_{\pm} = \frac{2\pi(n_c \pm \Delta n)}{L}, \quad a = \frac{\varepsilon_c}{k_c}. \quad (34)$$

The sideband ansatz below follows equation (33) of Xu and Guyenne [4], which follows the high-order spectral construction of Dommermuth and Yue [7]. Their published calculation uses a numerically computed exact steady Stokes carrier. The present corpus instead uses the project's analytic fifth-order carrier and randomizes the parameters over the support stated below; it is not a reproduction of that single published case. Let (η_c, ξ_c) be the project's analytic fifth-order deep-water Stokes carrier whose first elevation harmonic has amplitude a . This is an explicit approximation to the numerically exact steady Stokes carrier used by Xu and Guyenne; the two objects are not identified. The implemented initial state is the equation-(33) sideband form

$$\begin{aligned} \eta_0 &= \eta_c + \rho a [\cos(k_- x + \phi) + \cos(k_+ x + \phi)], \\ \xi_0 &= \xi_c + \rho a \left[\sqrt{\frac{g}{k_-}} e^{k_- \eta_0} \sin(k_- x + \phi) + \sqrt{\frac{g}{k_+}} e^{k_+ \eta_0} \sin(k_+ x + \phi) \right], \end{aligned} \quad (35)$$

followed by removal of the spatial mean of ξ_0 . Both sidebands travel with the carrier. Each case records $(n_c, \varepsilon_c, \Delta n, \rho, \phi)$, and the numerical depth is fixed to $h = 5L/(2\pi)$, so that even the first Fourier mode has $kh = 5$. There are no empirical cross terms and no exceptional rewrite of a sampled mode pair.

The leading deep-water modulation-instability condition is the Benjamin–Feir criterion [6]

$$0 < \frac{\Delta n/n_c}{2\sqrt{2}\varepsilon_c} < 1. \quad (36)$$

The declared support is

$$\begin{aligned} n_c &\in \{4, \dots, 20\}, & \varepsilon_c &\in [0.05, 0.13], \\ 1 \leq \Delta n < n_c, & \rho &\in [0.05, 0.20], & \phi &\in \mathbb{R}/(2\pi\mathbb{Z}), \end{aligned} \quad (37)$$

subject to (36). There are 66 feasible integer pairs $(n_c, \Delta n)$. For each family–split accepted-case quota, the production schedule assigns either $\lfloor C/66 \rfloor$ or $\lceil C/66 \rceil$ cases to each pair, where C is that quota; the first remainder categories in the frozen ordering receive the larger count. A rejected attempt is redrawn in the same pair category. Conditional on $(n_c, \Delta n)$, define

$$\varepsilon_{\min}(n_c, \Delta n) = \max\left\{0.05, \frac{\Delta n}{2\sqrt{2}n_c}\right\}. \quad (38)$$

The continuous draws are

$$\varepsilon_c \sim \text{Unif}(\varepsilon_{\min}, 0.13], \quad \rho \sim \text{Unif}[0.05, 0.20), \quad \phi \sim \text{Unif}[0, 2\pi). \quad (39)$$

These laws depend only on the sampled pair and not on a rollout outcome. The implemented population sampler treats the 66 pairs as parameter categories. In a 20,000-case schedule, the quotient–remainder rule gives 303 or 304 accepted cases to every pair. A separate 8448-case stress calculation drew 128 conditional samples in every category, and every sample satisfied (37) and (36). The ordered adapter stores the realized pair and continuous parameters before calling the constructor.

The Xu–Guyenne diagnostic point is

$$n_c = 9, \quad \Delta n = 2, \quad \varepsilon_c = 0.13, \quad \rho = 0.1, \quad \phi = -\frac{\pi}{4}. \quad (40)$$

It is retained as a named diagnostic configuration [4]. Using linear sidebands in this construction does not create a separate Airy training family: the complete modulated state is one Benjamin–Feir case.

4.4 JONSWAP/TMA random-phase states

The random-sea family uses the JONSWAP frequency spectrum rather than the Gaussian spectrum in the historical v9 generator. JONSWAP denotes the Joint North Sea Wave Project spectrum [8], and TMA denotes its standard Texel–Marsen–Arsloe finite-depth modification [9]. Choose a depth $h > 0$, peak wavenumber $k_p > 0$, and peak-enhancement factor $\gamma \geq 1$. For $k > 0$, define the finite-depth dispersion relation and group velocity by

$$\omega(k, h) = \sqrt{gk \tanh(kh)}, \quad c_g(k, h) = \frac{\partial \omega}{\partial k} = \frac{g[\tanh(kh) + kh \operatorname{sech}^2(kh)]}{2\omega(k, h)}. \quad (41)$$

Set the peak frequency $\omega_p = \omega(k_p, h)$, and let $k(\omega; h)$ denote the positive inverse of this dispersion relation. For angular frequency $\omega > 0$, define

$$\sigma(\omega) = \begin{cases} 0.07, & \omega \leq \omega_p, \\ 0.09, & \omega > \omega_p, \end{cases} \quad (42)$$

Define the unnormalized JONSWAP density directly by

$$\tilde{S}_J(\omega) = g^2 \omega^{-5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^{\exp[-(\omega - \omega_p)^2 / (2\sigma(\omega)^2 \omega_p^2)]}. \quad (43)$$

For a dimensionless depth–wavenumber product ζ , define the TMA factor and finite-depth spectrum by

$$\Phi_{\text{TMA}}(\zeta) = \frac{\tanh^2 \zeta}{1 + 2\zeta / \sinh(2\zeta)}, \quad \tilde{S}_T(\omega; h) = \tilde{S}_J(\omega) \Phi_{\text{TMA}}(k(\omega; h)h). \quad (44)$$

This factor tends to one in deep water and behaves like $\zeta^2/2$ as $\zeta \rightarrow 0$ [9]. Merely using finite-depth dispersion in the η -to- ξ transfer, as the historical Gaussian-sea generator did, is not a TMA elevation spectrum. The cited works supply the JONSWAP and TMA spectral forms. The Fourier-cell integration, fixed spectral window, direction split, phase sampling, and parameter strata below are choices of the replacement corpus.

Let $\Delta k = 2\pi/L$ and use the positive-mode set $\mathcal{J}_K$ from (4). Thus K remains a physical wavenumber cutoff rather than a mode index. To avoid point-sampling a narrow spectral peak, integrate the continuous spectrum over Fourier cells. For each $j \in \mathcal{J}_K$, define its energy E_j directly by

$$E_j = \int_{k_j - \Delta k/2}^{k_j + \Delta k/2} \tilde{S}_T(\omega(k, h); h) c_g(k, h) W_K(k) \, dk, \quad (45)$$

where the implementation evaluates each cell integral by fixed 16-point Gauss–Legendre quadrature. Define the normalized modal energy fractions

$$\varpi_j = \frac{E_j}{\sum_{\ell \in \mathcal{J}_K} E_\ell}, \quad \sum_{j \in \mathcal{J}_K} \varpi_j = 1. \quad (46)$$

The factor $c_g = \partial\omega/\partial k$ is the Jacobian from frequency density to wavenumber density. The parameter k_p is the wavenumber obtained by mapping the JONSWAP frequency-peak parameter through the finite-depth dispersion relation. TMA, cell integration, and W_K can shift the maximum of the discrete weights, so the realized maximizing mode is stored rather than assumed to equal k_p .

Let $H_s > 0$ denote the ensemble significant wave height and set $m_0 = (H_s/4)^2$. Let $r_d \in [0, 1]$ be the right-moving fraction in the traveling-wave decomposition of the linear Hamiltonian. For a positive scalar wavenumber, write $G_0(k; h) = k \tanh(kh)$ for the corresponding flat-surface DNO multiplier. Set

$$A_{j,+} = \sqrt{2r_d m_0 \varpi_j}, \quad A_{j,-} = \sqrt{2(1 - r_d) m_0 \varpi_j}. \quad (47)$$

For independent stored phases $\phi_{j,\pm}$, first define

$$\eta_0(x) = \sum_{j \in \mathcal{J}_K} [A_{j,+} \cos(k_j x + \phi_{j,+}) + A_{j,-} \cos(k_j x + \phi_{j,-})], \quad (48)$$

$$\xi_0(x) = \sum_{j \in \mathcal{J}_K} \frac{\omega(k_j, h)}{G_0(k_j; h)} [A_{j,+} \sin(k_j x + \phi_{j,+}) - A_{j,-} \sin(k_j x + \phi_{j,-})]. \quad (49)$$

Because right- and left-moving components can occupy the same spatial mode, the realized surface variance contains phase-dependent cross terms. Thus $H_s = 4\sqrt{m_0}$ is the spectral or ensemble significant height, not an exact identity for every realization. We do not rescale a realization to force that identity: when counterpropagating surface components nearly cancel, such a rescaling can make the kinetic energy arbitrarily large. Define the linear Hamiltonian H_{lin} by the following phase-independent expression:

$$H_{\text{lin}} = \frac{1}{2} \int_{\mathbb{T}_L} (\xi_0 G_0(h) \xi_0 + g \eta_0^2) dx = g L m_0, \quad (50)$$

and r_d is exactly the right-moving fraction of this linear energy. The realized $\|P_0^\perp \eta_0\|_{2,L}$, $\|\xi_0\|_{2,L}$, and interference ratio $4\|P_0^\perp \eta_0\|_{2,L}/H_s$ are recorded and included in the initial-geometry audit.

Let ϕ_+ and ϕ_- denote the arrays of right- and left-moving phases. Each random-sea case records $(h, H_s, k_p, \gamma, r_d, \phi_+, \phi_-)$. Storing the phase arrays is the most direct choice. A counter-based key is also sufficient only if the key-to-phase rule is versioned, indexed by physical mode, and stable across grid resolution.

The resolved-band window W_K equals one on the well-resolved interior and decays to zero at K . Its formula and transition wavenumber are common numerical configuration, not tunable case parameters. The constructor pilot freezes $K_0 = 96$ and $K = 128$ on the domain $L = 2\pi$. This placement follows directly from the declared support $k_p \leq 24$. Indeed, for $r \geq 1$,

$$\frac{\omega(rk_p, h)}{\omega(k_p, h)} = \left[r \frac{\tanh(rk_p h)}{\tanh(k_p h)} \right]^{1/2} \geq \sqrt{r}.$$

Thus $K_0 = 96 = 4 \max k_p$ leaves every frequency through $2\omega_p$ untapered, while the remaining 32 wavenumber cells provide the smooth transition to $K = 128$. Because this multiplication changes the tail of the physical spectrum, the result is called a *resolved-band, windowed JONSWAP/TMA spectrum*, not an exact untruncated TMA spectrum. This window is distinct from the common target projection P_K .

Increasing N at fixed L raises the largest resolved wavenumber but does not reduce the Fourier-cell width $2\pi/L$. The number of cells above half the largest cell energy is therefore recorded as a resolution diagnostic, not used to accept or reject a realization.

Sampling the dimensional variables independently is particularly dangerous in shallow water. Define instead

$$\chi_p = k_p h, \quad \delta_s = \frac{H_s}{2h}, \quad \varepsilon_p = \frac{k_p H_s}{2} = \chi_p \delta_s. \quad (51)$$

The quantity δ_s is a half-significant-height-to-depth descriptor; it is not the realized maximum crest height divided by depth. The corpus uses three strata of the same random-sea formula:

Table 1: JONSWAP/TMA training strata. The bounds define attempted cases; acceptance is reported separately.

Stratum	Attempted support
Shallow TMA	Reference-peak index n_p with $k_p = 2\pi n_p/L$ and $n_p \in \{16, 18, 20, 22, 24\}$, $\chi_p \in [0.2, 1.5]$, $h = \chi_p/k_p$, $\delta_s \in [0.03, 0.16]$, and $\chi_p \delta_s \leq 0.15$. For $L = 2\pi$, this gives $h \in [0.0083, 0.0938]$ and covers the diagnosed neighborhood $h \simeq 0.038$, $k_p h \simeq 0.7$.

Stratum	Attempted support
Finite-depth TMA	Draw $k_p \in [2, 12]$, $h \in [0.1, 1.5]$, and $H_s \in [0.005, 0.03]$.
Deep JONSWAP	Draw $k_p \in [2, 12]$, $h \in [5, 25]$, and $H_s \in [0.005, 0.03]$. Here the TMA factor is effectively one near the reference peak.

All three strata balance the peak-enhancement values $\gamma \in \{1, 3.3, 5\}$ and the right-moving fractions $r_d \in \{0, 1/2, 1\}$. Thus the production design has 27 parameter categories: one depth stratum, one value of γ , and one value of r_d . A requested family-split case count is divided among these categories by quotient and remainder, so their accepted-case quotas differ by at most one.

Within a shallow category, draw n_p uniformly from $\{16, 18, 20, 22, 24\}$. Draw χ_p uniformly on $[0.2, 1.5]$ and δ_s uniformly on $[0.03, 0.16]$, repeating this two-variable draw until $\chi_p \delta_s \leq 0.15$; equivalently, (χ_p, δ_s) is uniform with respect to area on the admissible part of that rectangle. Then set $k_p = 2\pi n_p / L$, $h = \chi_p / k_p$, and $H_s = 2h\delta_s$. Within a finite-depth or deep category, draw k_p , h , and H_s independently and uniformly on the three intervals in Table 1. In every category, all entries of the two phase arrays are independent and uniform on $[0, 2\pi)$. These population laws are implemented for all 27 categories.

All pseudorandom draws use deterministic, seeded streams produced by NumPy’s PCG64 generator. The durable proposal record stores the root seed, physical-family and revision identifiers, stream and attempt indices, every realized parameter, and both phase arrays. Before time integration, the common mathematical requirements are finiteness and the graph-domain condition $h + \eta_0(x) > 0$. The complete trajectory must still pass the common fixed-step numerical calculation in Section 6.

A deterministic 27-case pilot used nine cases in each stratum and every pair $(\gamma, r_d) \in \{1, 3.3, 5\} \times \{0, \frac{1}{2}, 1\}$ once per stratum. All 27 states were finite and in support. The largest relative error in the exact linear-energy identity (50) was 1.97×10^{-15} . The largest fixed-band $N = 512$ versus $N = 1024$ DNO discrepancy was 4.11×10^{-4} in shallow water, 7.76×10^{-5} at finite depth, and 6.07×10^{-5} in deep water. A short GL2-step check with $\delta = 0.01$ and 0.005 on the steepest member of each stratum had maximum discrepancy 3.85×10^{-8} . These are constructor and short-time checks; the full-horizon calculation is reported separately. The constructor returns both realized phase arrays. The pilot field archive did not store those arrays; its JSON record stores a per-case seed from which the present implementation replays them. The production record stores the arrays themselves, so replay does not depend on an unrecorded random-number call order.

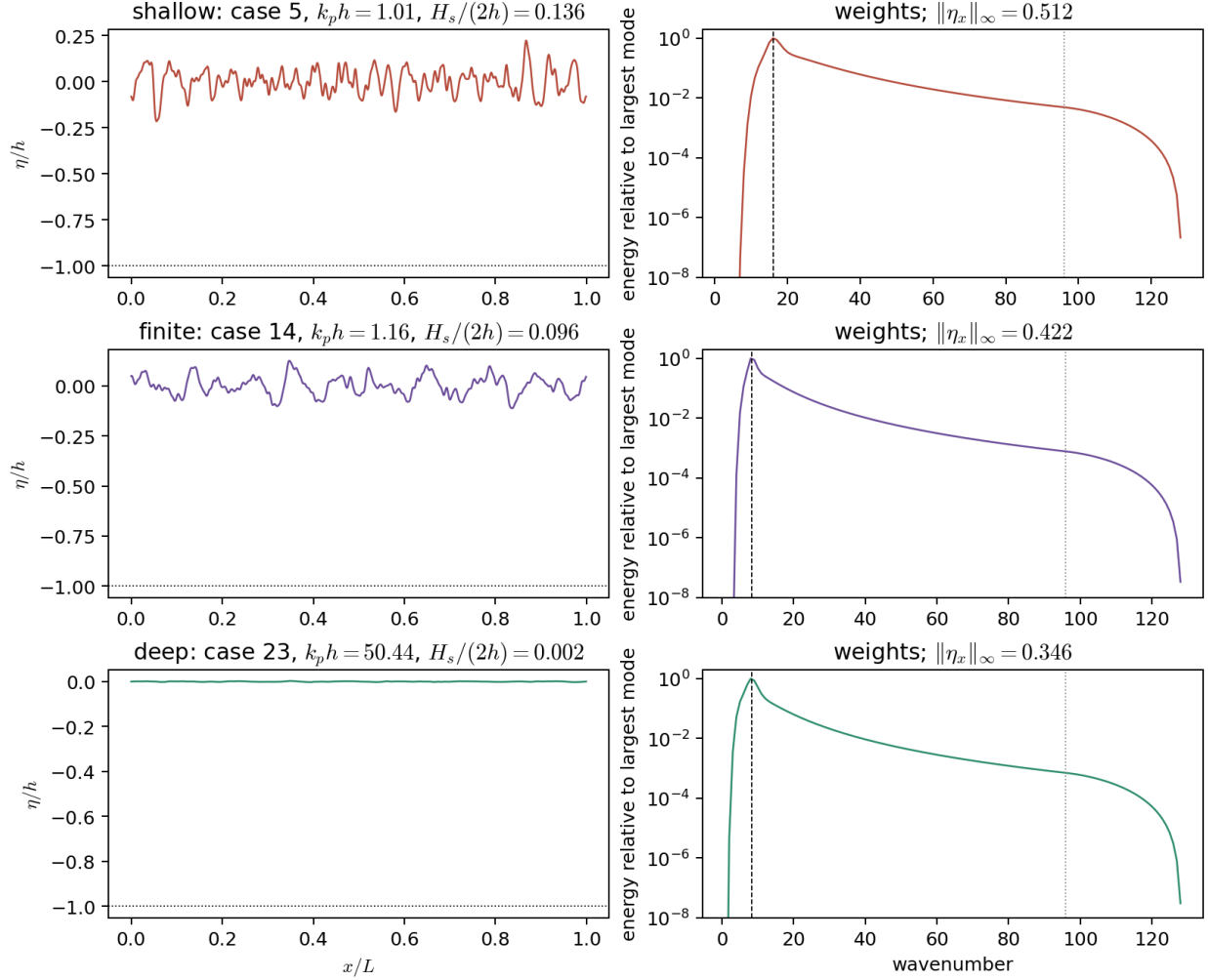


Figure 2: Representative stress cases from the resolved-band JONSWAP/TMA constructor pilot. One case is shown from each depth stratum. These cases are selected by predeclared parameter stress.

The shallow stratum replaces a role previously played by the first sparse packet addition. It does not automatically reproduce that data. Its roughness statistics and held-out shallow rollout accuracy must pass the decision gate in Section 9. The design also does not silently claim the old packet minimum $h = 0.005$: extending below 0.0083 requires a separate resolved-peak and reference-convergence study.

Equations (48)–(49) define JONSWAP/TMA-shaped random-phase Zakharov states. Because ξ_0 uses the linear branch relation and contains no bound-wave correction, they should not be called fully nonlinear developed seas. Their role is controlled, physically interpretable DNO coverage.

4.5 Production parameter sampling

Each family first receives an accepted-case count C in each data split. Suppose the family has m ordered parameter categories and write

$$C = bm + r, \quad 0 \leq r < m. \quad (52)$$

The first r categories receive $b + 1$ accepted cases and the other categories receive b . Let $B_i(C)$ denote this cumulative quota for category i . An additive training chunk that begins at C_0 cases per family and adds A cases requests

$$B_i(C_0 + A) - B_i(C_0) \quad (53)$$

new accepted cases from category i . Balancing each chunk independently would repeatedly favor the same remainder categories and would not produce a balanced nested learning curve.

Each family paragraph states the continuous law inside each category. All four samplers use deterministic PCG64 streams. Before construction or numerical integration, each attempted case stores its split, parameter category, root seed, stream identifier, and realized parameters. If the attempt is rejected, its record remains in the manifest and the replacement is drawn from the same category.

The replacement loop is finite. If category i requests $Q_i > 0$ accepted cases, it permits at most

$$A_i^{\max} = 4Q_i \quad (54)$$

durable case attempts. One attempt is one assigned case, not an internal Stokes amplitude redraw or a production GL2 rollout. A proposal reserves its attempt once, even if execution is interrupted. Pending work at the ceiling is replayed. If all $A_i^{\max}$ attempts have resolved and fewer than Q_i cases were accepted, the family run terminates and cannot enter a combined dataset. Thus the factor four is a computational bound, not a realized-field rejection rule. When the quota completes before the bound, it does not change the sampled or accepted cases.

4.6 Durable generation and dataset assembly

Each attempt is written in three possible stages:

proposal record $\longrightarrow$ accepted field file, when applicable $\longrightarrow$ result record.

The proposal is written first and contains the split, parameter category, random-stream coordinates, and complete sampled specification. The result record is the commit marker. If execution stops after the proposal or after the field file, restarting replays the same attempt rather than drawing a replacement. A result that names a missing field file, or a file with a different SHA-256 digest, is treated as corruption.

The run identity binds the requested quota, family, split, batch size, platform, random stream, numerical settings, software versions, dependency files, and generator source files. A change in any of these items cannot silently resume the same output directory. The combined dataset builder checks every included record and file, requires complete and gap-free nested chunks with equal accepted family counts in each split, and writes a compact row-to-case index. It does not copy the field arrays.

4.7 Training corpus at a glance

Table 2 summarizes the replacement corpus. It assigns the same accepted-case count to each family and balances the listed coarse parameter ranges within each family.

Table 2: Prospective four-family training corpus. “Times” counts stored states from a complete admissible fixed-step trajectory.

Family	Balanced parameter categories	Reference horizon and stored times	Case weight
Stokes	Finite/deep analytic branches; low and moderate categories from (16) within the current amplitude support, with finite-depth draws satisfying (20).	Static; one state per case.	1/4
Tanaka	Eleven (regime, m , q) categories, where q is the number of right-moving crests; amplitudes, centers, and separations recorded explicitly.	Final time 200; 200 times selected from the accepted trajectory using the surface-gradient activity in (81).	1/4
Benjamin–Feir	Deep-water carrier mode and steepness, one symmetric sideband separation, one sideband ratio, and one common phase from (37)–(36).	Final time 200; 200 times selected from the accepted trajectory using the maximum-elevation activity in (82).	1/4
JONSWAP/TMA	Shallow, finite, and deep strata from Table 1; γ , direction fraction, peak depth, relative height, and phase categories balanced explicitly.	Reference horizon $H = 16(2\pi/\omega_p)$; realized endpoint $T_s = .08\lfloor H/.08 \rfloor \leq H$; 16 approximately uniform saved times including both endpoints.	1/4

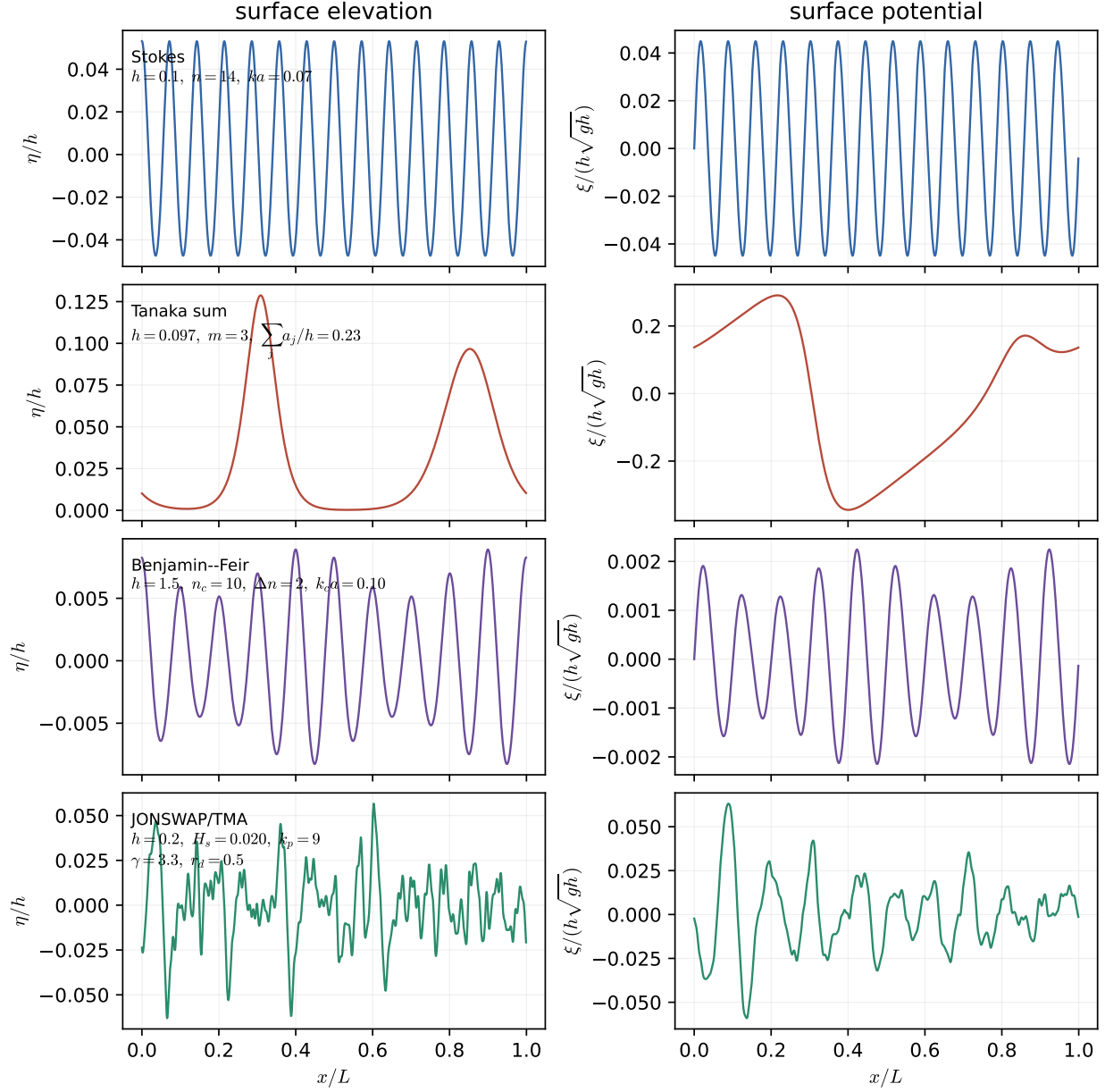


Figure 3: One constructed initial condition from each replacement training family. The columns show the dimensionless elevation η/h , surface potential $\xi/(h\sqrt{gh})$. The horizontal coordinate is x/L , and the numerical plots print selected scalar descriptors in the first column. The Stokes, Benjamin-Feir, and random-sea rows are generated directly from the formulas in this section. The random-sea illustration uses the frozen spectral window that starts at wavenumber 96 and vanishes at wavenumber 128. The Tanaka row is the fixed held-out case numbered 90000101 and was chosen without using model error.

Within the JONSWAP/TMA family the design gives equal case weight to its three depth strata. Within the other families the listed coarse strata are likewise balanced before seeded pseudorandom sampling within each category. In particular, the eleven Tanaka direction-count categories are balanced as evenly as each finite quota permits, so their counts differ by at most one. This gives additional aggregate weight to two- and three-crest states because they have more direction compositions. This is a transparent coverage design, not a claim that all categories are equally common in nature. If C is the accepted training-case count per family selected by the learning curve, the four-family training archive contains $C + 200C + 200C + 16C = 417C$ stored states before any storage-level compression. Thus the planned learning-curve grid $C \in \{2048, 4096, 8192, 16384\}$ corresponds to 854,016 through 6,832,128 training rows, while retaining $4C$ accepted training-case units. Validation, test, and rejected attempted cases are counted separately. The first checkpoint uses $C = 2048$ training cases per family and 1024 validation and 1024 test cases per family. It contains exactly $417(2048 + 1024 + 1024) = 1,708,032$ retained rows. With the current uncompressed writer, the field archives and row map are estimated to occupy 19.62 GiB; the current eager loader is estimated to peak at 39.6 GiB. The production target is $C = 16384$, with validation and test unchanged. Its 7,686,144 rows are close to the 7,633,989 rows in historical v9: 52,155 more rows, or 0.68%. The corresponding storage estimates are approximately 88.3 GiB of archives and row map and 178 GiB peak memory for the eager loader. Thus $C = 2048$ is a cumulative learning-curve checkpoint, not a completed replacement corpus. Rejected cases are not replaced by copying nearby accepted cases; the attempted and accepted counts are both reported. Using a fixed number of peak periods prevents shallow categories from facing a much longer acceptance test merely because their dimensional time scale differs. The historical packet *training generator* stored trajectories to final time 120. That archive does not set either the new packet diagnostic horizon of 20 time units or the random-sea training horizon.

5 Attempted and accepted counts

The tables above specify attempted populations. For every family and every declared parameter category, report both attempted and accepted counts. A category with no accepted cases is not filled from another category. Instead, generation stops and the support or numerical method must be revised explicitly.

The random-sea spectral window is already frozen at wavenumbers 96–128. Its 864-case configuration audit retained at least 98.469% of the reference spectral mass in the shallow stratum and at least 99.653% in the finite and deep strata. This is method-level evidence for the fixed constructor, not an individual-case test. The remaining population gate is computational: before bulk generation, run one accepted case from each of the 11 Tanaka, 66 Benjamin–Feir, and 27 random-sea parameter categories with the exact contract and the bounded scheduler. This is a 104-category trajectory pilot.

6 Reference evolution and whole-case acceptance

6.1 Discrete Zakharov flow

The reference trajectory is the solution of one declared discrete vector field, not merely the output of the two-stage, fourth-order Gauss–Legendre method (GL2). The depth h remains fixed. For $w = (\eta, \xi)$, define the internal order-six DNO evaluation and its delivered projection by

$$G_{\text{int}}(\eta; h)\xi = \mathcal{G}_M^{N,p}(P_K\eta; h)(P_K\xi), \quad G_{\text{ref}}(\eta; h)\xi = P_K^\circ[G_{\text{int}}(\eta; h)\xi]. \quad (55)$$

The nonlinear product is formed with $G_{\text{int}}(\eta; h)\xi$ before the common output projection. The discrete vector field is

$$f_h(w) = \begin{bmatrix} G_{\text{ref}}(\eta; h)\xi \\ P_K^\circ \left[-g\eta - \frac{1}{2}\xi_x^2 + \frac{(G_{\text{int}}(\eta; h)\xi + \eta_x\xi_x)^2}{2(1 + \eta_x^2)} \right] \end{bmatrix}. \quad (56)$$

Let $w_0 = (\eta_0, \xi_0)$ be an admissible candidate initial state, and write $z_0 = (\eta_0, \xi_0, h)$ when the fixed depth is also needed. The reference ODE is

$$\frac{dw}{dt} = f_h(w), \quad w(0) = w_0. \quad (57)$$

The products in the second component are evaluated on the common dealiased grid before projection. We integrate this ODE with the GL2 scheme in the next subsection. Removing the mean of $\partial_t \xi$ is precisely the time-dependent additive-potential gauge; it preserves $\int_{\mathbb{T}_L} \xi \, dx = 0$ without changing the velocity field or the surface evolution. The constructor projects both initial fields to $|k| \leq 128$. Inside every right-hand-side evaluation, both nonlinear components are projected to that same band, with the mean additionally removed from the ξ component as in (5). These projections are part of f_h . After every completed GL2 step, both state components are projected once more; that post-step projection is part of the numerical integrator rather than the ODE vector field.

For the integrating factor, define the linear part L_h and the remainder r_h by

$$L_h(\eta, \xi) = (G_0(h)\xi, -gP_0^\perp \eta), \quad r_h(w) = f_h(w) - L_h w. \quad (58)$$

On each nonzero Fourier mode, define $\omega_n^2 = g|k_n| \tanh(h|k_n|)$. The linear evolution is the explicit rotation

$$\begin{bmatrix} \widehat{\eta}_n(t) \\ \widehat{\xi}_n(t) \end{bmatrix} = \begin{bmatrix} \cos(\omega_n t) & \frac{|k_n| \tanh(h|k_n|)}{\omega_n} \sin(\omega_n t) \\ -\frac{g}{\omega_n} \sin(\omega_n t) & \cos(\omega_n t) \end{bmatrix} \begin{bmatrix} \widehat{\eta}_n(0) \\ \widehat{\xi}_n(0) \end{bmatrix}. \quad (59)$$

The zero modes are unchanged. We write this linear evolution as e^{tL_h} . With $v(t) = e^{-tL_h}w(t)$,

$$\dot{v} = \widetilde{F}(t, v) := e^{-tL_h}r_h(e^{tL_h}v). \quad (60)$$

6.2 Residual-controlled Gauss–Legendre stages

The two-stage, fourth-order Gauss–Legendre coefficients are

$$c_{1,2} = \frac{1}{2} \mp \frac{\sqrt{3}}{6}, \quad A = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \end{bmatrix}. \quad (61)$$

Let t_n and v_n be the current time and integrating-factor state, and let $\delta_n > 0$ be the step length. Write a_{ij} for the entries of the matrix A in (61). The two stage values satisfy

$$V_i = v_n + \delta_n \sum_{j=1}^2 a_{ij} \widetilde{F}(t_n + c_j \delta_n, V_j), \quad i = 1, 2. \quad (62)$$

After the stages converge,

$$v_{n+1} = v_n + \frac{\delta_n}{2} \sum_{i=1}^2 \widetilde{F}(t_n + c_i \delta_n, V_i). \quad (63)$$

For a candidate pair (V_1, V_2) , define the fixed-point maps

$$\Phi_i(V_1, V_2) = v_n + \delta_n \sum_{j=1}^2 a_{ij} \tilde{F}(t_n + c_j \delta_n, V_j), \quad i = 1, 2. \quad (64)$$

Write V_i^η, V_i^ξ and Φ_i^η, Φ_i^ξ for their two Fourier-space components, and let $\|\cdot\|_{\ell^2}$ be the Euclidean norm of one component's Fourier coefficients. The implemented relative stage defect is

$$R_n = \max_{\substack{i \in \{1,2\} \\ u \in \{\eta, \xi\}}} \frac{\|\Phi_i^u - V_i^u\|_{\ell^2}}{\max\{\|\Phi_i^u\|_{\ell^2}, \|V_i^u\|_{\ell^2}, \text{tiny}\}}, \quad (65)$$

where tiny is the smallest positive normal number of the working floating-point type. Normalizing η and ξ separately avoids adding fields with different units. By Parseval's identity, the Fourier normalization cancels from each ratio.

The stage tolerance τ_{GL} and iteration cap are fixed in the generator specification. A stage is declared converged only when $R_n \leq \tau_{\text{GL}}$; otherwise its iteration ends at the declared cap or when a nonfinite stage is detected. Xu and Guyenne report that two to four iterations typically achieved relative errors between 10^{-6} and 10^{-8} in their applications [4]; that observation motivates a tolerance study, not an unchecked iteration count. The implementation recomputes (65) from the returned stages, freezes each batched sample after convergence, and records finite cap exhaustion separately from a nonfinite stage or state. With $\tau_{\text{GL}} = 10^{-8}$ and an eight-iteration cap, a four-code-path $N = 256$ smoke converged all 32 sampled steps in two updates. The largest residuals for Airy, finite Stokes, Tanaka, and the pre-revision BF constructor were respectively 9.08×10^{-11} , 1.10×10^{-10} , 4.63×10^{-10} , and 2.09×10^{-9} .

6.3 Method-level fixed-band resolution validation

This validation is performed on a predeclared panel before production and does not accept or reject individual corpus cases. Resolution is tested by reconstructing one physical specification, not by interpolating one sampled array. Fix a complete case record and re-evaluate the same family formula with the same stored parameters on the three grids $N_r = 2^r N$, $r = 0, 1, 2$. Each reconstruction is performed independently in float64. Write η_r and ξ_r for the fields reconstructed on grid N_r . Define the dimensionless fields

$$U_r^{(\eta)} = \frac{\eta_r}{h}, \quad U_r^{(\xi)} = \frac{\xi_r}{h\sqrt{gh}},$$

and write $\widehat{U}_{r,n}^{(\nu)}$ for the Fourier coefficient of $U_r^{(\nu)}$, where $\nu \in \{\eta, \xi\}$. The initial-state discrepancy is

$$\Delta_{\text{IC},K} = \max_{\nu \in \{\eta, \xi\}} \max_{0 \leq r < s \leq 2} \frac{\left(\sum_{|k_n| \leq K} |\widehat{U}_{r,n}^{(\nu)} - \widehat{U}_{s,n}^{(\nu)}|^2 \right)^{1/2}}{\left(\sum_{|k_n| \leq K} |\widehat{U}_{s,n}^{(\nu)}|^2 \right)^{1/2} + 10^{-30}}. \quad (66)$$

On grid N_r , compute the target field

$$G_r^\star = P_K^\circ \left[\mathcal{G}_M^{N_r, p}(P_K \eta_r; h)(P_K \xi_r) \right], \quad (67)$$

set $U_r^{(G)} = G_r^\star / \sqrt{gh}$, and write $\widehat{U}_{r,n}^{(G)}$ for its Fourier coefficients. Then

$$\Delta_{G,K} = \max_{0 \leq r < s \leq 2} \frac{\left(\sum_{|k_n| \leq K} |\widehat{U}_{r,n}^{(G)} - \widehat{U}_{s,n}^{(G)}|^2 \right)^{1/2}}{\left(\sum_{|k_n| \leq K} |\widehat{U}_{s,n}^{(G)}|^2 \right)^{1/2} + 10^{-30}}. \quad (68)$$

The frozen DNO threshold for the stratified configuration-validation panel is $\Delta_{G,K} \leq 10^{-3}$. Both statistics are evaluated on a predeclared, family-stratified panel before a generator revision is frozen. A failed panel identifies an unresolved constructor or an unsupported parameter category; it does not authorize deleting individual production rows by inspecting their shape. Thus the expensive order- M calculation is not repeated on three grids at every stored time.

The historical Tanaka generator instead used a rowwise sign-transition rule. If $D_j = [G_{\text{old}}(\eta; h)\xi](x_j)$ is the sampled old target, that rule formed the cyclic forward difference $d_j = (D_{j+1} - D_j)/\Delta x$, discarded entries below $0.03 \max_j |d_j|$, and rejected when the remaining cyclic sign sequence had more than ten transitions. It was useful for the old clipped-profile constructor because a boundary seam generated oscillatory curvature, but a resolved physical wave may also have many extrema. In the completed Tanaka calibration at the historical band $K = 341$, all 12 matched clipped cases had $\Delta_{G,K} \geq 2.33 \times 10^{-2}$; the same specifications rebuilt by the periodic constructor had $\Delta_{G,K} \leq 1.10 \times 10^{-5}$, and four clean controls had $\Delta_{G,K} \leq 1.64 \times 10^{-4}$. This supports the mechanism without claiming that the current corpus was already filtered by the three-grid statistic. The tangent-Hermite construction removes the seam mechanism, the three-grid panel checks its spatial resolution before production, and production applies no sign-transition rejection.

6.4 Method-level selection of the production time step

Two different intervals were called **dt** in earlier project notes. Let Δs denote the spacing between saved states and let δ denote the actual GL2 step in (62)–(63). If one saved interval is divided into m equal GL2 steps, then

$$\delta = \frac{\Delta s}{m}. \quad (69)$$

Production uses

$$\Delta s = 0.08, \quad m = 8, \quad \delta = 0.01. \quad (70)$$

Thus 0.08 is a storage interval, not the step of the GL2 method.

The choice $\delta = 0.01$ was made before the present corpus plan. The May 2026 time-step study used $m = 8$ while varying the saved interval through $\Delta s \in \{0.01, 0.02, 0.04, 0.08, 0.16, 0.32\}$. In terms of the actual GL2 step, the tested values were therefore

$$\delta \in \{0.00125, 0.0025, 0.005, 0.01, 0.02, 0.04\}.$$

Four Tanaka cases were integrated to $T = 50$. Relative terminal elevation differences from the $\delta = 0.00125$ calculation were about 10^{-11} , 10^{-10} , and 10^{-9} at $\delta = 0.0025$, 0.005 , and 0.01 , respectively. The difference jumped to 6.2×10^{-2} at $\delta = 0.02$. The historical statement “use **dt**=0.08” therefore selected the largest successful *saved interval with eight substeps*; it selected the actual GL2 step $\delta = 0.01$, not $\delta = 0.08$.

Xu and Guyenne use the same GL2 step 0.01 in their reported calculations [4]. Their paper does not determine our saved spacing, residual tolerance, or complete-case policy. Those are numerical choices for this corpus.

The step choice was checked again in July 2026 on twelve predeclared full-horizon cases from the trajectory families. For this method-level validation only, each initial condition was integrated with $\delta = 0.01$ and 0.005 and compared at common saved times. At a saved time t_j , define

$$Y_\delta^{(\eta)}(t_j) = \frac{P_K \eta_\delta(t_j)}{h}, \quad Y_\delta^{(\xi)}(t_j) = \frac{P_K \xi_\delta(t_j)}{h\sqrt{gh}}, \quad Y_\delta^{(G)}(t_j) = \frac{G_{\text{ref}}(\eta_\delta(t_j); h)\xi_\delta(t_j)}{\sqrt{gh}}, \quad (71)$$

where $(\eta_\delta, \xi_\delta)$ is the trajectory computed with GL2 step δ . Let $\mathcal{A} = \{\eta, \xi, G\}$ label the three displayed fields. For a case specification θ , the validation discrepancy was

$$E(\theta; \delta) = \max_{\nu \in \mathcal{A}} \frac{\max_j \left\| Y_{\delta/2}^{(\nu)}(t_j) - Y_\delta^{(\nu)}(t_j) \right\|_{2,L}}{\max_j \left\| Y_{\delta/2}^{(\nu)}(t_j) \right\|_{2,L} + 10^{-12}}. \quad (72)$$

The maximum over time in the denominator prevents a temporary cancellation of one field from producing a large relative error. Of the twelve cases, ten produced complete trajectories under both steps, and all ten satisfied

$$E(\theta; 0.01) \leq 1.126337 \times 10^{-5}.$$

The remaining steep Tanaka and equation-(33) Benjamin–Feir cases became incomplete at nearly the same physical times under both steps. Halving the step therefore neither changed an acceptance decision nor rescued either calculation, and no 0.0025 calculation was needed. Together with the May study, this panel validates the common step $\delta = 0.01$. It does not justify repeating the half-step calculation for every production case. On this panel the reported integration times were 4026.06 seconds for the 0.01 calculations and 7798.47 seconds for the 0.005 calculations. The pair therefore used 11824.53 seconds of integration work, whereas the 0.01 arm alone used 34.0% of that total. Repeating the validation pair in production would nearly triple the measured work of the selected method.

6.5 Whole-case production acceptance

Production consequently performs one rollout per attempted trajectory case, with $\delta = 0.01$, and stores candidate states every $\Delta s = 0.08$. After the initial condition has passed its declared family support, there is one whole-trajectory acceptance condition:

The fixed numerical method must return a complete admissible trajectory on the requested interval $[0, T]$.

Here T is the horizon declared by the case. “Complete admissible trajectory” means that every scheduled GL2 step reaches T , every implicit stage satisfies $R_n \leq \tau_{\text{GL}} = 10^{-8}$, every saved state is finite and satisfies $h + \eta > 0$ on the numerical grid, and every delivered target $G_{\text{ref}}(\eta; h)\xi$ at a saved time is finite. Failure to reach T , a nonfinite field or target, an unsolved stage, and loss of the water-wave graph domain are diagnostic causes of this one failure condition; they are not separate fitted filters. If the condition fails, the complete case is rejected and no finite prefix enters the corpus. There is no per-case 0.005 comparison, discrepancy threshold, or 0.0025 retry.

For a batch of B initial conditions, the production executor advances all B cases once with $\delta = 0.01$. The eight GL2 steps between consecutive saved times are carried out inside this one batched calculation. Random-sea cases in the same batch can have different declared horizons. The batch is advanced to its longest horizon, but each case and its stage record are restricted to that case’s declared saved-time prefix before completeness and admissibility are decided. Values beyond a shorter case’s horizon do not belong to that case.

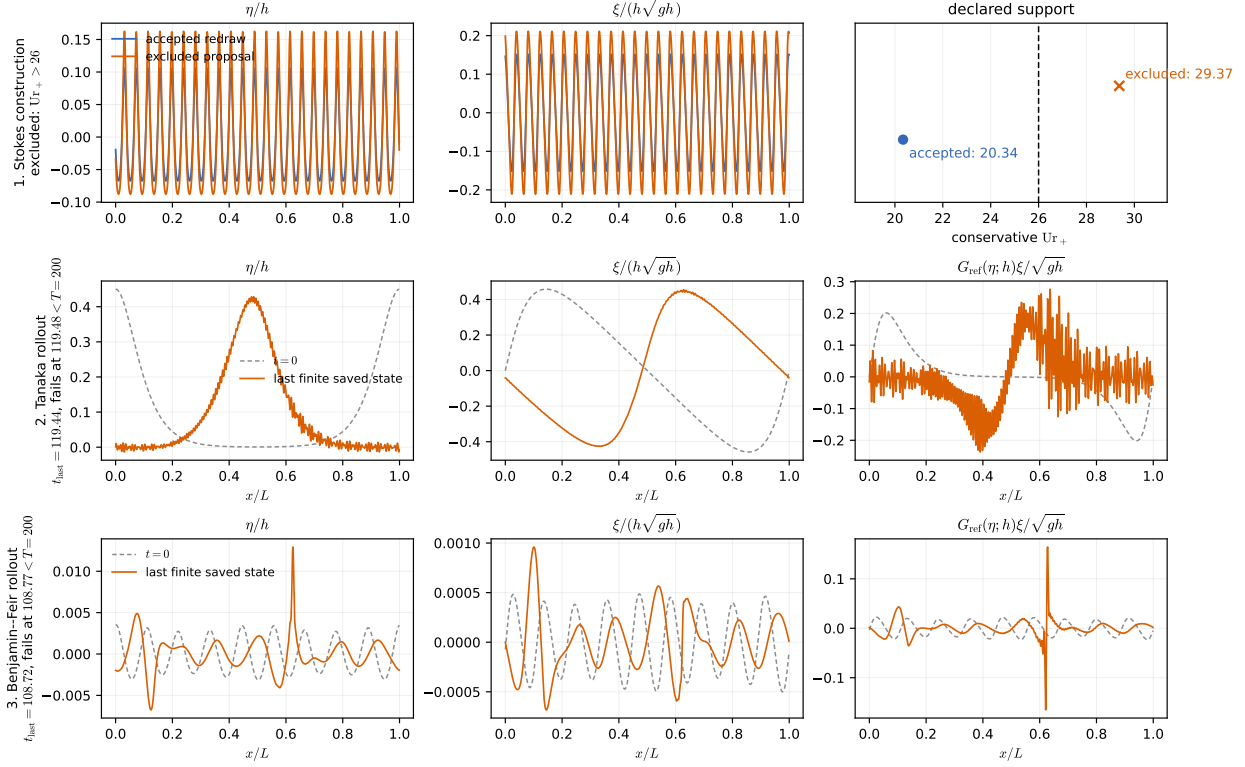


Figure 4: Examples of the two current decisions. In the top row, current finite-depth Stokes validation attempt 2 first proposes an amplitude with $Ur_+ = 29.37 > 26$. That amplitude is outside the declared Stokes support. With the carrier, depth, phase, and steepness category unchanged, the next amplitude has $Ur_+ = 20.34$ and is accepted. Both displayed profiles are finite; the decision is the stated support inequality, not their appearance. The middle and bottom rows use only the $\delta = 0.01$ arms of two full-horizon validation cases that lie in their declared family supports. The Tanaka trajectory has its last finite saved state at $t = 119.44$ and first failed GL2 stage at $t = 119.48$. The Benjamin–Feir trajectory has its last finite saved state at $t = 108.72$; its stage at $t = 108.77$ reaches eight iterations with residual $3.8931 \times 10^{-7} > 10^{-8}$. Since both request $T = 200$, the fixed method returns no complete trajectory and each case would contribute zero rows. These are fixed validation examples, not estimates of production rejection probabilities. No half-step calculation enters this figure or either decision.

Static Stokes states are not trajectories and do not use the complete-trajectory condition. A static case is retained only when its declared support produces a finite state with positive water column and a finite common target. The three-grid, cross-family spatial audit in the preceding subsection instead validates the constructor and common production grid before generation; it is not a case-level filter.

The implicit stage tolerance $\tau_{\text{GL}} = 10^{-8}$ remains part of the definition of the GL2 algorithm in the preceding subsection. It specifies when a stage equation has been solved, just as a linear-solver tolerance does; it is not a second data-quality classifier. Likewise, the only water-column condition used here is the mathematical graph requirement $h + \eta > 0$. The earlier half-depth margin $h + \eta \geq h/2$ is not part of the contract.

6.6 Diagnostics not used for acceptance

For grid index ℓ , write $\eta_\ell = \eta(x_\ell)$ and $\xi_\ell = \xi(x_\ell)$. The monitored discrete energy is computed in float64 using the unprojected order- M DNO output evaluated from the internal filtered state:

$$\mathcal{H}_M^N(z) = \frac{\Delta x}{2} \sum_{\ell=0}^{N-1} \left[\xi_\ell (\mathcal{G}_M^{N,p}(\eta; h) \xi)_\ell + g \eta_\ell^2 \right], \quad (73)$$

not reconstructed from the stored float32 $G_{\text{ref}}(\eta; h) \xi = P_K^\circ [\mathcal{G}_M^{N,p}(P_K \eta; h)(P_K \xi)]$ target. Because the implemented vector field includes spectral filtering and a truncated DNO, $\mathcal{H}_M^N$ is not assumed to be an exact invariant of that discrete evolution; its drift is a numerical diagnostic. Let z_n be the state after time step n , and set the fixed denominator floor $\varepsilon_{\mathcal{H}} = 10^{-14}$. The reported statistic is

$$D_{\mathcal{H}} = \max_n \frac{|\mathcal{H}_M^N(z_n) - \mathcal{H}_M^N(z_0)|}{|\mathcal{H}_M^N(z_0)| + \varepsilon_{\mathcal{H}} g h^2 L}. \quad (74)$$

It is reported on the predeclared numerical validation panels together with mass, spectra, and stage iteration counts to validate the common numerical method. These quantities are not required fields in every production shard. They do not enter case acceptance.

6.7 Storage after acceptance

Every requested storage time is a GL2 step endpoint: consecutive saved times are separated by eight steps of length 0.01. No temporal interpolation is used. Stored frames are selected only after the numerical method has returned a complete admissible trajectory, so a failed trajectory never contributes isolated training rows. The production step and saved spacing are fixed by (70); they are not adjusted case by case.

7 Reference-method and surrogate validation

Xu and Guyenne [4] organize validation around direct DNO tests, truncation order, dealiasing, long-time traveling waves, invariants, and time reversal. The replacement generator adopts the same separation of questions.

For two states $z = (\eta, \xi, h)$ and $\tilde{z} = (\tilde{\eta}, \tilde{\xi}, h)$ at the same depth, define the dimensionless state norm

$$\|(\eta, \xi)\|_h^2 = \|\eta/h\|_{2,L}^2 + \left\| \xi / (h\sqrt{gh}) \right\|_{2,L}^2, \quad (75)$$

where the normalized spatial norm $\|\cdot\|_{2,L}$ is defined in (3). Define their relative state error by

$$\text{rel}_h(z, \tilde{z}) = \frac{\|(\eta - \tilde{\eta}, \xi - \tilde{\xi})\|_h}{\|(\tilde{\eta}, \tilde{\xi})\|_h + 10^{-12}}. \quad (76)$$

For a validation step $\delta > 0$, let z_δ and $z_{\delta/2}$ denote trajectories sampled at common comparison times t_ℓ , where ℓ indexes those times. Define the method-level temporal defect by

$$E_t(\delta) = \max_\ell \text{rel}_h(z_\delta(t_\ell), z_{\delta/2}(t_\ell)). \quad (77)$$

For a traveling profile, let $\sigma \in [0, L]$ denote a horizontal translation and define

$$E_{\text{shape}}(t) = \min_{\sigma \in [0, L]} \frac{\|\eta(\cdot, t) - \eta(\cdot - \sigma, 0)\|_{2,L}}{\|\eta(\cdot, 0)\|_{2,L} + 10^{-12}}, \quad (78)$$

and obtain the numerical displacement from the minimizing σ . Starting from z_0 , integrate to time T and call the result z_f . Starting from z_f , use the same step lengths in reverse order with negative sign and call the returned state z_b . The forward-backward defect is

$$E_{\text{rev}}(T) = \text{rel}_h(z_b, z_0). \quad (79)$$

Thus the forward and backward calculations use exactly the same step lengths.

Successive- M differences on general states are called order sensitivity, not convergence to the physical DNO, unless an independent boundary-integral or conformal reference is supplied. This qualification matters because Xu–Guyenne report an amplitude- and resolution-dependent optimal order: larger M need not reduce error once the recursion is ill-conditioned.

The completed three-grid Tanaka study remains useful evidence that periodic wrapping removes the old clipped-profile defect. It does not by itself validate the four training families, the three diagnostic suites, the time integrator, or the choice $M = 6$.

8 Temporal sampling, splitting, and normalization

Temporal selection is performed only after the fixed-step calculation has returned a complete admissible trajectory. Tanaka and Benjamin–Feir trajectories are evaluated on the dense time grid

$$t_j = 0.08j, \quad j = 0, \dots, 2500, \quad (80)$$

and 200 times are retained separately for each case. The internal GL2 step is 0.01; 0.08 is the spacing of the candidate storage grid.

For Tanaka, define the discrete surface-gradient energy and its activity by

$$S_{T,j} = \sum_{\ell=0}^{N-1} |\partial_x \eta(x_\ell, t_j)|^2, \quad A_{T,j} = \frac{|\delta S_{T,j}|}{S_{T,j} + 10^{-12}}. \quad (81)$$

Here

$$\delta S_0 = S_1 - S_0, \quad \delta S_j = \frac{S_{j+1} - S_{j-1}}{2} \quad (1 \leq j \leq 2499), \quad \delta S_{2500} = S_{2500} - S_{2499}.$$

Table 3: Validation objects, measured quantities, and decision roles.

Panel	Quantity reported	Role
Manufactured DNO	Exact-DNO error, discrete-target error, learned-target error, and end-to-end error as defined in Appendix B.2.	Selects the discrete reference configuration, not network weights.
Three-grid construction	$\Delta_{\text{IC},K}$ and $\Delta_{G,K}$ from (66)–(68).	Required before production; validates each constructor and the common spatial configuration.
Time-step study	$E_t(\delta)$, maximum stage residual, and $D_{\mathcal{H}}$.	Required before production; selects one common GL2 step. It is a method-level panel, not a calculation repeated for each corpus case.
Traveling Stokes/Tanaka	$E_{\text{shape}}(t)$, displacement divided by t , mass, (74), and $E_{\text{rev}}(T)$.	Validates long-time propagation of the reference method.
Benjamin–Feir	Modal amplitudes $2 \hat{\eta}_n(t) $ at the carrier and sidebands, recurrence error, $D_{\mathcal{H}}$, and high-band energy.	Validates modulation dynamics; growth alone is not labeled breaking.
Airy	Flat-surface identity and cubic small-amplitude remainder scaling.	Post-freeze architecture verification only.
Sparse Fourier	One-step DNO error and complete rollout error; nonfinite rollouts count as infinite error.	Development cases select only the corpus; final cases only report.
Support sensitivity	Cellwise attempted/accepted counts under predeclared neighboring family-support bounds.	Tests whether a declared support bound removes a difficult parameter category; it does not alter the fixed production step.

The derivative $\partial_x \eta$ is computed by the Fourier multiplier ik_n on the common grid. Thus Tanaka receives extra samples when the spatial surface derivative changes rapidly during crest interaction.

For Benjamin–Feir, define instead

$$S_{B,j} = \max_{0 \leq \ell < N} |\eta(x_\ell, t_j)|, \quad A_{B,j} = \left(S_{B,j} - \min_r S_{B,r} \right)^4. \quad (82)$$

The carrier dominates $\sum_\ell |\eta_x|^2$ and can hide the slow modulation, so Benjamin–Feir uses the maximum absolute surface elevation rather than the surface-gradient signal.

For either family, convolve its nonnegative activity values with a normalized discrete Gaussian, extending the endpoint values outside the time interval. The Gaussian standard deviation is 50 candidate-time steps for Tanaka and 20 for Benjamin–Feir; its discrete support is truncated at three standard deviations, rounded upward to an integer number of time indices. Call the smoothed values $\tilde{A}_{T,j}$ and $\tilde{A}_{B,j}$. For $F \in \{\text{T}, \text{B}\}$, define

$$p_{F,j} = \frac{1 - \alpha_F}{2501} + \alpha_F \frac{\tilde{A}_{F,j}}{\sum_{r=0}^{2500} \tilde{A}_{F,r}}, \quad \alpha_{\text{T}} = 0.5, \quad \alpha_{\text{B}} = 0.85. \quad (83)$$

If the smoothed activity vanishes identically, its normalized term is replaced by the uniform distribution. The 200 retained indices are the deterministic inverse-CDF locations of the midpoint quantiles $(\ell + 1/2)/200$, $\ell = 0, \dots, 199$. The first and last candidate times are included, repeated indices are advanced to the next unused index, and all selected indices and activity parameters are stored with the case. This rule depends only on the accepted reference trajectory, never on model error.

For a random-sea trajectory, let

$$T_{\text{hor}} = 16 \frac{2\pi}{\omega_p}, \quad T_s = 0.08 \left\lfloor \frac{T_{\text{hor}}}{0.08} \right\rfloor, \quad J_s = \frac{T_s}{0.08}.$$

It retains the saved-grid indices

$$j_\ell = \left\lfloor \frac{\ell J_s}{15} + \frac{1}{2} \right\rfloor, \quad t_\ell = 0.08 j_\ell, \quad \ell = 0, \dots, 15.$$

Thus the integration ends at the last saved-grid point not exceeding sixteen peak periods, and the retained endpoints are 0 and T_s . Static Stokes cases retain only $t = 0$, with phase already contained in the case record.

Random-sea cases in one numerical batch can have different values of T_s . The single production batch is advanced to the longest required saved-grid endpoint in the batch, but each case's fields and GL2 stage record are cropped to its own declared prefix before completeness and admissibility are decided. Hence values after a shorter case's horizon cannot affect its acceptance. A regression test obtains the same decision, retained fields, and prefix stage record in serial and batched runs even when post-horizon values are deliberately made nonfinite. The all-category GPU pilot uses batch size four and will determine whether that size is retained for production.

Training, validation, and test cases come from three disjoint tagged PCG64 construction streams. Their root seeds are respectively 2026072210, 2026072204, and 2026072205. Every attempted case stores its root seed, stream identifier, and realized parameters. Let s be the split root, let f identify the physical family, let v identify the generator revision, let r be the stream identifier, and let a be the attempted-case index. Every family initializes its per-case PCG64 stream from the five integer words (s, f, v, r, a) . Thus replay of one attempted case reproduces its complete specification without depending on the number of earlier accepted cases. Split membership is therefore fixed when the attempted case specification is written, before integration or time selection; it is not assigned from the resulting trajectory. The training stream continues until it has the post-acceptance count C per family selected by the learning curve. The validation and test streams each contain 1,024 accepted cases per family, balanced over the same parameter categories to within one case. Every rejected attempt remains attached to its original split manifest. A nested training chunk is never enlarged in place. The successive (C_0, A) pairs, where C_0 is the preceding cumulative count and A is the new count per family, are

$$(0, 2048), (2048, 2048), (4096, 4096), (8192, 8192).$$

They use distinct stream identifiers 0, 1, 2, 3 and the category increments in (53). Validation and test each use one fixed, non-nested chunk. A requested count in one family-split category is never filled by copying or silently drawing from another category. In the static Stokes generator, one candidate fixes the mode, depth, phase, and steepness category and permits at most 1000 amplitude redraws within that same category. The sampler record contains every rejected amplitude and its Ur_+ value, followed by the accepted amplitude on success. If all 1000 redraws are exhausted, the sampler raises with the complete strict record. The implemented static executor stores that record

as a zero-row out-of-support attempt and schedules a replacement in the same category. It is not a fatal batch error. An unclassified error propagates without being relabeled; only the trajectory executor’s explicit fatal exception writes a terminal failure record. This internal amplitude-redraw limit does not consume additional outer attempts. The replacement case does consume the next durable attempt under (54).

No two temporal snapshots from the same case can appear in different subsets. This removes exact temporal-sibling leakage; it does not imply that nearby parameter draws are statistically independent.

The loader implements the case-balanced rule stated in Section 1: it selects exactly one stored time from every accepted training case in every epoch and batches those selected rows. The validation time for each case is fixed deterministically. A different number of stored times requires a separate learning-curve comparison; it is not chosen from rollout failures.

Let $\mathcal{I}_{\text{train}}$ denote all stored rows from accepted training cases. For row $i \in \mathcal{I}_{\text{train}}$, let η_i and ξ_i be its two input fields, let h_i be its depth, and let $G_i^* = G_{\text{ref}}(\eta_i; h_i)\xi_i$ be its target from (12). The network retains the scale normalization used by the strongest current checkpoint, but estimates it on this set only:

$$s_\eta = \max_{i \in \mathcal{I}_{\text{train}}, x} |\eta_i(x)|, \quad s_\xi = \max_{i \in \mathcal{I}_{\text{train}}, x} |\xi_i(x)|, \quad s_G = \max_{i \in \mathcal{I}_{\text{train}}, x} |G_i^*(x)|. \quad (84)$$

Let $h_\star = L/(2\pi)$ be the fixed reference length. For one row, write (η, ξ, h, G^*) for its two input fields, depth, and target. The network receives

$$\tilde{\eta} = \eta/s_\eta, \quad \tilde{\xi} = \xi/s_\xi, \quad d_h = \log\left(\frac{h}{h_\star}\right), \quad \tilde{G}^* = G^*/s_G. \quad (85)$$

Hence d_h is dimensionless and is unchanged by the period rescaling in Appendix C; on the production domain $L = 2\pi$ it has the same numerical value as the existing input $\log h$. The intended corpus gives positive values for all three scales, and the release audit verifies this before training. The current loader nevertheless uses one as a defensive fallback if a nonpositive scale reaches it; the writeup does not count that fallback as validation of a malformed corpus. No per-family, per-mode, validation, or diagnostic statistic enters (84). The cached scales are bound to the checked identity of the assembled dataset. Reports include both the number of case units and the number of stored snapshots. Corpus size is selected by a learning curve in accepted case count, rather than by reproducing the v9 snapshot count by default. The Airy, manufactured, and sparse diagnostic cases are absent from all three training subsets and from the training-scale calculation. If a diagnostic result motivates a redesign, the final paper reports a fresh, case-disjoint diagnostic draw after the revised model is frozen.

At C accepted training cases per family, the case-balanced loader presents $4C$ samples per epoch. With global batch size 1024, comparisons at a fixed epoch count would therefore give larger corpora more optimizer updates. The learning curve instead uses 10,240 optimizer updates:

C	Steps per epoch	Epochs for 10,240 updates
2,048	8	1,280
4,096	16	640
8,192	32	320
16,384	64	160

The learning-rate schedule is a function of optimizer step, not epoch. Checkpoints are compared at the same predeclared update numbers; the number of epoch boundaries is not allowed to give the smaller corpus more checkpoint-selection opportunities. Test errors are first aggregated by case within each family and then with equal weight across the four families. A single statistic weighted by the 1, 200, 200, and 16 stored rows per case is not a paper-level four-family result.

9 Execution plan

The target, four population laws, spatial window, reference integrator, complete-trajectory condition, storage rule, and bounded quota mechanism are frozen. Before the production executor was simplified to one time step, an exact-contract four-family smoke used run-specification schema 1 and computed the 0.01/0.005 validation pair:

Family	Accepted	Rows	Pair discrepancy	Wall time (s)
Stokes	1	1	—	3.782
Tanaka	1	200	3.1953×10^{-8}	5,555.35
Benjamin–Feir	1	200	3.7058×10^{-8}	5,561.11
JONSWAP/TMA	1	16	7.6600×10^{-7}	835.19

The four cases assembled into one checked dataset view containing 417 finite rows on 1024 points, and the loader selected one time from each case. The pair discrepancies are method-validation evidence, and the recorded wall times include work that production no longer performs. The numerical target, constructors, GL2 method, field writer, view, and loader did not change when the attempt bound was added, but the scheduler source did. Consequently these run-spec-schema-1 artifacts deliberately cannot resume as a run-spec-schema-2 run.

The eleven-category Tanaka law advances the global generator revision to 2, and its stored parameter record uses the corresponding version-2 Tanaka schema. Revision-1 parameter smokes, dry runs, and all prior family pilot artifacts remain useful as historical constructor and transaction evidence, but they do not satisfy the revision-2 run specification and cannot be resumed into a revision-2 run. The 104-category trajectory gate and the entire final corpus use revision 2.

Current generator-revision-2 numerical evidence consists of one exact case from each of the four Stokes categories. All four were accepted, their quality-decision masks were zero, and the complete calculation took 4.6875 seconds. The old reduced real-GL2 transactions from the trajectory families check historical software wiring only; they are not revision-2 trajectory evidence. The single-arm production executor and writer pass all 31 focused acceptance, method-audit, production-integration, writing, and quota tests; the launcher and reduced real-GL2 smoke pass all six focused tests. The remaining work is:

1. **Complete the running all-category GPU gate.** This pilot is separate from the corpus and contributes no training, validation, or test rows. Under the exact run-spec-schema-2 and generator-revision-2 contract, request one accepted case from each of the 11 Tanaka, 66 Benjamin–Feir, and 27 random-sea parameter categories, using batch size four and the ceiling of four attempts per category. The resulting trajectory gate contains 104 cases. It passes only if every category accepts one case, every run is complete with no pending work or exhausted ceiling, all required arrays and numerical decisions are finite, every rejection has a decoded support reason or a diagnostic cause of an incomplete or inadmissible fixed-step trajectory, the three runs have compatible numerical and software identities, and wall time and peak GPU memory are recorded.
2. **Generate the first checkpoint.** Generate 2048 accepted training cases and 1024 accepted validation and test cases from each family. A category-ceiling failure terminates its family run; it is not repaired by borrowing cases from another category. The expected retained count is 1,708,032 rows.
3. **Build and audit one dataset view.** Include all twelve completed family–split summaries, verify their records and file hashes, and build the row-to-case index without copying fields.

Check the exact family, split, case, and row counts, one-time-per-case training and validation selections, finite arrays, disjoint full case identities (family, revision, split, stream, attempt), and normalization scales computed only from all stored training rows.

4. **Run the complete nested learning curve.** Extend the training split additively to

$$C \in \{2048, 4096, 8192, 16384\}$$

accepted cases per family while keeping validation and test fixed. The final point contains 7,686,144 retained rows, comparable to historical v9’s 7,633,989 snapshots. Train two seeds at every C with the same architecture, optimizer, and 10,240-update schedule. Compare final checkpoints at that common update count. Report every point rather than stopping at an undefined “material improvement” threshold.

5. **Test shallow mixed-mode coverage.** Evaluate every learning-curve endpoint on the 11 historical shallow-steep cases and the frozen sparse development panel. The existing checkpoint C27, trained on all v9 sources, is an external performance baseline, not a matched ablation. Measure relative terminal elevation error and assign infinite error to every nonfinite rollout. Report the median, 95th percentile, nonfinite count, and counts above 0.25, 0.50, 0.75, and 1. A failure motivates a separately versioned population revision; it does not change this frozen corpus after results are seen.
6. **Freeze and test once.** After all models and checkpoints are fixed, evaluate the untouched training-family test cases, Airy panel, manufactured-DNO panel, and final sparse panel. Report errors by case within family and then weight the four family summaries equally. Any later design change requires a new final sparse draw.

10 Claims this plan does and does not support

If completed, the plan supports the following statements:

- every attempted initial condition is reproducible from a complete, finite-dimensional case specification and one common configuration;
- every training label uses the same discrete operator convention;
- training, validation, and test rows come from disjoint cases;
- no Airy, manufactured-harmonic, or sparse-Fourier diagnostic row contributes to optimization or empirical normalization;
- every saved state of an accepted reference trajectory remains in the water-wave graph domain, and the trajectory is complete under the declared fixed GL2 method; and
- the declared four-family depth, nonlinearity, and spectral coverage is reported in physical dimensionless variables.

It does not, without further analysis, support the stronger statements that the selected G_M equals the physical DNO, that Hamiltonian conservation proves trajectory accuracy, that JONSWAP spans every physically relevant sea, or that a trajectory rejected by this numerical method is physically impossible. Success on the four training families also does not establish arbitrary mixed-mode generalization or accuracy below the declared shallow-TMA depth; those remain empirical questions measured by the diagnostic suites. The existing C27 result cannot answer the corpus-ablation question because C27 was trained on all v9 sources. The order, grid, timestep, benchmark, and new-corpus retraining studies are included precisely to keep those questions separate.

A Audit of the current v9 archive

The archive `combined_dataset_v9.npz` contains 7,633,989 snapshots on $N = 1024$ points over $L = 2\pi$. Its six construction categories are Gaussian seas (460,740 rows), manufactured single modes (500,000), fifth-order Stokes states (1,000,000), Tanaka-profile states (2,200,000), Benjamin–Feir states (2,033,249), and sparse Fourier packets (1,440,000). The suffixes `g0`, `g1`, and `modal` name generator runs, not different initial-condition families or DNO orders.

The archive remains a valuable empirical baseline. It was assembled from generators with different row counts, filtering rules, and target conventions. The relevant distinctions are summarized in Table 4.

Table 4: Why the flat v9 archive does not define the replacement paper corpus.

Object	Current archive	Replacement rule
Sampling rule	Source-dependent numbers of stored times make snapshot weight differ from case weight.	In every epoch select one stored time from every accepted case, then batch those selected rows.
Discrete target	Static sources retain the direct order-six output, whereas rollout sources are projected to $ k \leq 128$.	Use (12) for every row.
Acceptance	Early sources were filtered rowwise; later sources used source-specific whole-case Hamiltonian tests.	Use declared support for static Stokes states and the one common complete fixed-step trajectory condition for the other three families.
Reconstructibility	Some phase arrays and case fields were not stored, and no stage residual was recorded.	Store every parameter and test the residual (65).
Statistical split	A rowwise split can place temporal siblings in different subsets; normalization used the complete archive.	Split by attempted case and fit scales on accepted training cases.
Support changes	Some failed parameter combinations were replaced by hand, for example one Benjamin–Feir mode pair.	Declare support first; record rejection without changing parameters.

Table 5 records the historical parameter envelopes needed for the direct comparison. These are support bounds, not empirical quantiles, and independent dimensional draws did not induce uniform density in the dimensionless coordinates used below. Here H_s is significant wave height, k is wavenumber, k_p is peak wavenumber, k_c is carrier wavenumber, b is spectral bandwidth, and n , n_c , and n_j are Fourier mode indices. The symbols a and a_j are wave amplitudes, Δn is sideband separation, and $\rho_{\pm}$ are sideband-to-carrier amplitude ratios.

Two historical formulas matter for interpreting the redesign. First, the old sea uses the following Gaussian elevation spectrum rather than the JONSWAP/TMA construction in Section 4.4:

$$S_{\text{old}}(k; k_p, b) = \exp \left[-\frac{(|k| - k_p)^2}{2(k_p/b)^2} \right]. \quad (86)$$

Second, the source archived as *linear* is a manufactured harmonic trace. For wavenumber k , amplitude a , phase ϕ , and direction sign $s \in \{-1, 1\}$, set $\vartheta = kx + \phi$ and $\omega^2 = gk \tanh(kh)$. Then

$$\eta(x) = a \cos \vartheta, \quad (87)$$

$$\xi(x) = s \frac{ag \cosh(k(h + \eta(x)))}{\omega \cosh(kh)} \sin \vartheta. \quad (88)$$

Table 5: Historical support envelopes retained for comparison.

Construction	Declared support in v9
Gaussian sea	$H_s \in [0.005, 0.03]$, $k_p \in [2, 12]$, bandwidth $b \in [1.5, 5]$; $h \in [0.1, 1.5]$ or $h \in [5, 25]$.
Manufactured mode	$n \in \{1, \dots, 20\}$, $h \in [0.02, 1.5]$, $kh \in [0.02, 20]$, and $0.005 \leq ka \leq 0.15$, subject to $a \in [10^{-4}, 0.011494]$.
Stokes	Deep: $n \in \{1, \dots, 20\}$, $h \in [4, 50]$. Finite: $n \in \{14, \dots, 26\}$, $h \in [0.02, 1.5]$, $kh \in [0.5, 5]$. Both impose $ka \leq 0.15$.
Tanaka	Main: $h \in [0.01, 0.30]$, one to three crests, total $\sum_j a_j/h \in [0.10, 0.35]$. Steep: $h \in [0.20, 0.35]$, one crest, $a/h \in [0.25, 0.45]$.
Benjamin–Feir	$h \in [0.5, 4]$, $n_c \in \{4, \dots, 20\}$, $k_c a \in [0.05, 0.13]$, $\Delta n \in \{2, 3, 4\}$, and $\rho_{\pm} \in [0.05, 0.20]$.
Sparse Fourier	One to five modes, $h \in [0.005, 0.5]$, $kh \in [0.2, 1.5]$, total $a/h \in [0.03, 0.25]$, and $\sum_j n_j a_j \leq 0.15$.

The nonlinear dependence on η makes this an exact manufactured boundary trace, not a separate physical linear-wave family.

The historical depth union is approximately $0.005 \leq h \leq 50$. This is not automatically the support of the replacement dataset. The four-family design reaches $h \simeq 0.0083$ through its shallow random-sea stratum and $h = 50$ through Stokes; the smaller-depth interval remains diagnostic unless a separate resolution study admits it. The full source-by-source audit is therefore evidence for the redesign, not the main definition of it.

B Diagnostic specifications and separation from training

The three diagnostic suites are Airy, exact manufactured, and sparse Fourier cases. They have complete case specifications, but they are not used for training. The sparse-Fourier cases are split into disjoint development and final-test sets. Sparse development cases may reject or revise the replacement corpus through the predeclared gates in Section 9; they never contribute loss terms or normalization statistics. Final-test cases are generated from new case keys and remain untouched until the corpus, network, optimizer, and checkpoint are frozen. If a final-test result causes another design change, it ceases to be final and another case-disjoint test draw is required.

The Airy grid is a fixed analytic verification panel, not a random development/test draw, and is evaluated only after the network is frozen. The manufactured panel is separate: it may select the reference discretization (M, N, p, K) because its purpose is to verify that numerical formula, but it never selects network parameters. No final diagnostic row contributes to optimization, empirical normalization, network-hyperparameter selection, or checkpoint selection.

B.1 Airy linear-limit panel

For the exact flat-surface test, choose an integer mode $n \geq 1$, set $k = k_n$, and choose a depth $h > 0$, trace amplitude $b_A > 0$, and phase $\psi \in [0, 2\pi)$. The identity is

$$\eta = 0, \quad \xi = b_A \cos(kx + \psi), \quad G(0; h)\xi = k \tanh(kh)\xi. \quad (89)$$

For paired phase-space tests, choose an elevation amplitude $a > 0$, phase $\phi \in [0, 2\pi)$, and direction sign $s \in \{-1, 1\}$. With $G_0(k; h) = k \tanh(kh)$ and $\omega(k, h) = \sqrt{gG_0(k; h)}$, define

$$\eta_A(x) = a \cos(kx + \phi), \quad (90)$$

$$\xi_A(x) = s a \frac{\omega(k, h)}{G_0(k; h)} \sin(kx + \phi). \quad (91)$$

These are valid Zakharov initial data but are not asserted to be exact nonlinear traveling waves at appreciable ka . Let $\mathcal{N}_A$ be the tested integer modes, let $\mathcal{X}_A$ contain the tested values of the dimensionless depth $\chi = kh$, and let $\mathcal{E}_A$ contain the tested values of the steepness $\epsilon = ka$:

$$\begin{aligned} \mathcal{N}_A &= \{2, 4, 8, 16, 24\}, \\ \mathcal{X}_A &= \{0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20\}, \\ \mathcal{E}_A &= \{10^{-4}, 10^{-3}, 10^{-2}, 0.03, 0.075, 0.15\}. \end{aligned} \quad (92)$$

For every $n \in \mathcal{N}_A$, $\chi \in \mathcal{X}_A$, and $\epsilon \in \mathcal{E}_A$, set

$$k_n = \frac{2\pi n}{L}, \quad h = \frac{\chi}{k_n}, \quad a = \frac{\epsilon}{k_n}, \quad (93)$$

and retain only categories satisfying

$$0.005 \leq h \leq 50, \quad \frac{a}{h} = \frac{\epsilon}{\chi} \leq 0.16. \quad (94)$$

Use both $s = \pm 1$ and phases $\phi \in \{0, \sqrt{2}\}$ radians; the latter are not related by an integer-grid shift on the $N = 1024$ grid. The flat test uses the same (n, χ) categories, $\psi \in \{0, \sqrt{2}\}$, and $b_A \in \{10^{-3}, 10^{-2}, 10^{-1}\}$. These rules determine (n, h, a, ϕ, s) or (n, h, b_A, ψ) completely and exclude bottom-crossing surfaces.

The amplitude ladder tests the declared small-surface structure of the *discrete target*; no independent nonlinear physical-DNO solver is assumed here. Define

$$\begin{aligned} r_A^*(a) &= G_{\text{ref}}(\eta_A; h) \xi_A \\ &\quad - P_K^o[G_0(h)P_K \xi_A + G_1(P_K \eta_A; h)P_K \xi_A]. \end{aligned} \quad (95)$$

Both η_A and ξ_A are proportional to a . Therefore $G_0(h)\xi_A$ has degree one in a , $G_1(\eta_A; h)\xi_A$ has degree two, and $G_m(\eta_A; h)\xi_A$ has degree $m + 1$. After subtracting the first two terms, the remainder therefore begins at degree three:

$$\|r_A^*(a)\|_{2,L} = O(a^3) \quad (a \rightarrow 0) \quad (96)$$

at fixed (k, h, ϕ, s) and fixed (M, N, p, K) . When the projected G_2 coefficient is nonzero, the predicted leading slope is three; a vanishing coefficient is reported rather than fitted on roundoff. The diagnostic is the least-squares slope of $\log \|r_A^*(a)\|_{2,L}$ against $\log a$ over the three smallest usable amplitudes. The learned remainder is defined by replacing $G_{\text{ref}}(\eta_A; h)\xi_A$ with the network output. Thus the flat-surface identity is a physical analytic test, whereas the nonlinear amplitude ladder tests consistency with the declared polynomial target and architecture. Thus this held-out panel is more informative about the analytic $G_0 + G_1 + O(\eta^2)$ architecture than a large collection of similar single-mode training rows.

B.2 Manufactured exact-DNO panel

For an integer mode $n \geq 1$, set $k = k_n$. Choose a depth $h > 0$, amplitude $a > 0$, phase $\phi \in [0, 2\pi)$, and direction sign $s \in \{-1, 1\}$, and set $\omega = \omega(k, h) = \sqrt{gk \tanh(kh)}$. Define

$$\varphi(x, y) = s \frac{ag \cosh(k(y+h))}{\omega \cosh(kh)} \sin(kx + \phi), \quad \eta(x) = a \cos(kx + \phi). \quad (97)$$

The boundary trace is exactly (88), and the normal derivative is known without solving a boundary-value problem:

$$[G_{\text{exact}}(\eta; h)\xi](x) = (\varphi_y - \eta_x \varphi_x)_{y=\eta(x)}. \quad (98)$$

This is the one-dimensional counterpart of the direct DNO test in Xu–Guyenne equations (30)–(32) [4]. For the exact benchmark, the full trace $\xi = \varphi|_{y=\eta}$ is evaluated on a sufficiently resolved grid and the unprojected $\mathcal{G}_M^{N,p}(\eta; h)\xi$ is compared with $G_{\text{exact}}(\eta; h)\xi$. This is distinct from the training target (12): in general

$$G(\eta; h)(P_K \xi) \neq G(\eta; h)\xi. \quad (99)$$

The frozen manufactured grid uses the same $n \in \mathcal{N}_A$, $kh \in \mathcal{X}_A$, $ka \in \mathcal{E}_A$, feasibility condition (94), phases $\phi \in \{0, \sqrt{2}\}$, and signs $s = \pm 1$ as the Airy panel. The constructor (97) then determines the full boundary trace and exact normal derivative, so every benchmark case is reconstructible.

Once the reference configuration is frozen, let G_{net} denote the frozen learned surrogate. Report four different errors:

$$e_{\text{CS}} = \frac{\|\mathcal{G}_M^{N,p}(\eta; h)\xi - G_{\text{exact}}(\eta; h)\xi\|_2}{\|G_{\text{exact}}(\eta; h)\xi\|_2}, \quad (100)$$

$$e_{\text{target}} = \frac{\|G_{\text{ref}}(\eta; h)\xi - P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}{\|P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}, \quad (101)$$

$$e_{\text{learn}} = \frac{\|G_{\text{net}}(P_K \eta; h)(P_K \xi) - G_{\text{ref}}(\eta; h)\xi\|_2}{\|G_{\text{ref}}(\eta; h)\xi\|_2}, \quad (102)$$

$$e_{\text{total}, K} = \frac{\|G_{\text{net}}(P_K \eta; h)(P_K \xi) - P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}{\|P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}. \quad (103)$$

Here e_{CS} measures the unprojected order- M reference, e_{target} is the total discrepancy of the declared discrete target and therefore combines Craig–Sulem, grid, input-projection, and output-projection effects; it is not an isolated projection error. e_{learn} isolates fitting of the declared discrete target, and $e_{\text{total}, K}$ is the end-to-end fixed-band error. This prevents reference truncation or projection error from being credited to or blamed on the network.

B.3 Sparse-Fourier shallow stress suite

These cases test shallow mixed-mode generalization and are never used for training. Choose a depth $h > 0$, a mode count $m \in \{2, 3, 4, 5\}$, and distinct positive Fourier indices $n_1, \dots, n_m$. For each component, choose an amplitude $a_j > 0$, phase $\phi_j \in [0, 2\pi)$, and direction $s_j \in \{-1, 1\}$. The initial state is

$$\eta_{\text{F}}(x) = \sum_{j=1}^m a_j \cos(k_{n_j} x + \phi_j), \quad (104)$$

$$\xi_{\text{F}}(x) = \sum_{j=1}^m s_j a_j \frac{\omega(k_{n_j}, h)}{G_0(k_{n_j}; h)} \sin(k_{n_j} x + \phi_j). \quad (105)$$

Here $G_0(k; h) = k \tanh(kh)$ and $\omega(k, h) = \sqrt{gG_0(k; h)}$, as in the Airy construction. Fresh cases satisfy

$$0.02 \leq h \leq 0.12, \quad 0.2 \leq k_{n_j} h \leq 1.5, \quad 0.03 \leq \sum_{j=1}^m \frac{a_j}{h} \leq 0.18, \quad \sum_{j=1}^m k_{n_j} a_j \leq 0.15. \quad (106)$$

Each fresh panel contains 256 attempted cases. Divide the depth interval logarithmically into four parts, divide the allowed total amplitude $\sum_j a_j/h$ into a lower and an upper half, and distinguish unidirectional cases, in which all s_j agree, from bidirectional cases, in which both signs occur. Attempt four cases in every combination of the four depth intervals, four mode counts, two amplitude halves, and two direction classes. A deterministic seeded PCG64 stream selects the remaining continuous quantities within each category. The stream identifier and all modes, amplitudes, phases, directions, case indices, and realized parameters are stored.

The development and final panels use the respective seeds 2026072201 and 2026072202. Rejection never causes replacement. At least two of the four attempts in every category must pass the common reference checks; otherwise the panel construction fails. The final panel remains untouched until the corpus, model, optimizer, and checkpoint rule are frozen.

The 11 historical shallow initial conditions are also retained as a separate development panel, but their trajectories are recomputed with the same frozen reference solver used everywhere else; the old trajectory files are historical evidence only. All sparse-Fourier rollouts have final time 20 and are stored every 0.08 time units. An earlier unmatched comparison suggested that shallow packets reduced the shallow error tail, but simultaneous changes in corpus size, normalization, and device count prevent a causal claim. That observation motivates this held-out test, not a sparse-Fourier training family.

C Rescaling to the 2π reference torus

The period dependence of G is suppressed in (9). In this appendix only, write $G^{[L]}(\eta; h)$ for the physical DNO on $\mathbb{T}_L$. Set $\lambda = L/(2\pi)$ and write $x = \lambda X$, $y = \lambda Y$. Define

$$\bar{\eta}(X) = \frac{\eta(\lambda X)}{\lambda}, \quad \bar{\xi}(X) = \xi(\lambda X), \quad \bar{h} = \frac{h}{\lambda}. \quad (107)$$

$$\boxed{[G^{[L]}(\eta; h)\xi](\lambda X) = \frac{1}{\lambda}[G^{[2\pi]}(\bar{\eta}; \bar{h})\bar{\xi}](X).} \quad (108)$$

Indeed, $\bar{\varphi}(X, Y) = \varphi(\lambda X, \lambda Y)$ solves the same Laplace–Neumann–Dirichlet problem below $\bar{\eta}$ at depth $\bar{h}$. The chain rule gives $\bar{\varphi}_X = \lambda\varphi_x$, $\bar{\varphi}_Y = \lambda\varphi_y$, and $\bar{\eta}_X(X) = \eta_x(\lambda X)$, which proves (108) directly.

The cutoff in this document is a physical wavenumber cutoff. Let $P_K^{[L]}$ denote projection on $\mathbb{T}_L$ onto integer Fourier indices $j \in \mathbb{Z}$ satisfying $|2\pi j/L| \leq K$, and let $P_{\lambda K}^{[2\pi]}$ denote the corresponding projection on $\mathbb{T}_{2\pi}$. For $\bar{f}(X) = f(\lambda X)$,

$$(P_K^{[L]}f)(\lambda X) = P_{\lambda K}^{[2\pi]}\bar{f}(X). \quad (109)$$

Write $G_{\text{ref}, K}^{[L]}$ for the operator in (12) evaluated on $\mathbb{T}_L$ with cutoff K , and write $G_{\text{ref}, \lambda K}^{[2\pi]}$ for the corresponding operator on $\mathbb{T}_{2\pi}$. If the same Fourier indices are present on both N -point grids, then, up to floating-point roundoff,

$$\boxed{\lambda[G_{\text{ref}, K}^{[L]}(\eta; h)\xi](\lambda X) = [G_{\text{ref}, \lambda K}^{[2\pi]}(\bar{\eta}; \bar{h})\bar{\xi}](X).} \quad (110)$$

For Fourier index j , $k_j = 2\pi j/L = j/\lambda$, whereas the corresponding reference-torus wavenumber is $\bar{k}_j = j$. Define $\bar{a} = a/\lambda$, $\bar{H}_s = H_s/\lambda$, and $\bar{k}_p = \lambda k_p$. Then $k_j h = \bar{k}_j \bar{h}$, $k_j a = \bar{k}_j \bar{a}$, $a/h = \bar{a}/\bar{h}$, $H_s/(2h) = \bar{H}_s/(2\bar{h})$, and $k_p H_s/2 = \bar{k}_p \bar{H}_s/2$. Thus each dimensionless combination used in the family definitions is unchanged by a uniform rescaling of length.

The production corpus remains on $L = 2\pi$. Equations (108)–(110) are pre-normalization identities for DNO evaluation; they do not by themselves define a variable-period training augmentation. A full trajectory rescaling at fixed gravity uses $\tau = t/\sqrt{\lambda}$. The rescaled fields are

$$\eta^\sharp(X, \tau) = \lambda^{-1} \eta(\lambda X, \sqrt{\lambda} \tau), \quad \xi^\sharp(X, \tau) = \lambda^{-3/2} \xi(\lambda X, \sqrt{\lambda} \tau),$$

$$[G^{[2\pi]}(\eta^\sharp(\cdot, \tau); \bar{h}) \xi^\sharp(\cdot, \tau)](X) = \lambda^{-1/2} [G^{[L]}(\eta(\cdot, \sqrt{\lambda} \tau); h) \xi(\cdot, \sqrt{\lambda} \tau)](\lambda X).$$

It must therefore be treated as a separate change of variables rather than an archive-level spatial resize.

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